

A
Major Project
On
**VULNERABILITY EVALUATION METHOD FOR E-
COMMERCE TRANSACTION SYSTEMS WITH
UNOBSERVABLE TRANSITIONS**

Submitted to
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in
Computer Science & Engineering (AI&ML)
by
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Under the Guidance Of
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Affiliated to JNTUH – Approved by AICTE – Accredited by NBA

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DEPARTMENT OF COMPUTER SCIENCE AND ENGINEERING



CERTIFICATE

This is to certify that the project entitled “**VULNERABILITY EVALUATION METHOD FOR E-COMMERCE TRANSACTION SYSTEM WITH SYSTEM WITH UNOBSERVABLE TRANSITIONS**” is submitted by **DASI SUSMITHA (216Y5A6601)**, in the partial fulfillment of requirement for the award of degree of Bachelor of Technology in Computer Science and Engineering(AI&ML) during academic year 2023-24.

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ABSTRACT

As the popularity of e-commerce grows, an electronic coupon (e-coupon) is widely used due to its convenience and portability. In most e-coupon services, the information of e-coupons is managed on a centralized server. However, e-coupon services are often vulnerable to security issues because of centralization. For example, when the e-coupon information which is stored in a centralized e-coupon server is forged, it becomes difficult to match the user and the e-coupons owner, and an expired e-coupon can be used repetitively (i.e., double-spending). To handle this issue, we propose a new e-coupon service by exploiting a blockchain system to improve the security of the service. To do this, we first design a server to enable the e-coupon service and communicate with the blockchain system. Second, we devise a smart contract on the blockchain system to provide integrity of the e-coupon business logic and the e-coupons information. We implemented the proposed service on an Ethereum-based blockchain system. The experimental results show that our proposed service improves higher security with a minor performance overhead compared with an existing e-coupon service.

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CHAPTER 1

INTRODUCTION

With the development of e-commerce, an increasing number of people are paying attention to the study of economic systems. Due to the imperfect nature of the e-commerce system itself, there are many problems in managing e-commerce systems. There are some useful modeling methods to describe e-commerce systems, such as those of , Du et al, use a labeled Petri net (LPN) to analyze the obligation and accountability of cooperative systems and extend the LPN to a labeled work flow net (LWN) to model e-commerce work flows. However, they do not consider the three parties (shopper, merchant, and third-party payment platform (TPP)) involved in the transaction. Yu et al, propose an e-commerce business process net (EBPN) to construct an e-commerce business process. Based on the EBPN, they describe malicious behavior patterns. The behavior patterns represent potential attacks that violate security. However, they are not suitable for vulnerable e-commerce systems with unobservable actions.

For a safe system, there are methods that can be used to diagnose insecurity for users, such as those, in fact, this premise of system security is idealized. Most of the time, an e-commerce system is not completely secure. Thus, it is important to analyze vulnerable e-commerce systems. For a vulnerable e-commerce system, our previous work proposes vulnerable e-commerce transaction nets (VET- nets). VET nets consider both normal actions and malicious actions. Malicious actions include attacks and unobservable actions. In this paper, we use VET-nets as modeling tools. For a VET- net, we know that if we provide a known attack form, we can locate the cause of the attack form in an e-commerce system using, this methods. Depending on the attack, there are vulnerable points used to complete the attack process. In fact, there may not be only one of these points. Some points are closely related to these unobservable transitions. These actions may play different roles in a VET-net. The different effects can be represented in terms of the effect degree. This effect degree is called the vulnerability degree. Existing methods mainly aim at observable system attacks, and they are suitable for vulnerability diagnosis from observable activities.

For some unobservable activities and unpredictable attack forms, these methods are not applicable. From the perspective of observed actions, one only

analyzes these unobservable states of malicious activity. Such a method ignores the effects of unobservable activities that lead to a malicious state. It cannot perform the appropriate adjustment of the e-commerce system. In other words, the system mostly performs analysis after malicious activity, and it does not prevent the effects. It is mostly postmortem analysis. It has difficulty preventing attacks.

Another method only considers unexpected attack forms and ignores unobservable actions. It is not good for testing the e-commerce system itself. So our goal is to address two issues (i.e., unknown attacks and unobservable actions). The starting point of security practices for e-commerce systems is to evaluate their vulnerability. When a potential threat exploits the vulnerable actions of the system, it can lead to the destruction and damage of the system. Vulnerability assessment is a process of interpretation and vulnerability analysis. The purpose of vulnerability assessment is to find and control vulnerability. In fact, if we can control these unobservable actions and assess the occurrence probability of each path to the final condition, we can diagnose and assess the e-commerce system. Therefore, we emphasize unobservable actions.

Our goal is to determine how to use a formal method to identify the vulnerable actions of a VET-net system and to give a formal calculation method for the vulnerability degree of a VET-net. The contributions of this paper include:

- We propose the concept of vulnerable transitions, which includes not only vulnerable actions but also unobservable transitions. We present a new method to locate the vulnerable transitions of a VET-net.
- We use hidden Markov models (HMMs) to approximate the state reach ability graph of a VET-net. Based on this, we give a vulnerable state evaluation method for a VET-net.

It shows the framework of our methods. In the first stage, by capturing the user transaction behaviors of the e-commerce trading process, we analyze the observable and unobservable actions and then construct a VET-net model. Based on the VET-net, we can find all paths to the final condition; then, according to the characteristics of behavior. In the third step, we can obtain a hidden Markov model according to the reach ability graph of the VET-net. According to the hidden Markov model and reach ability graph, we can compute the vulnerability degree of the VET-net.

CHAPTER 2

LITERATURE SURVEY

2.1 An empirical study of click fraud in mobile advertising networks:-

Smartphone advertisement is increasingly used among many applications and allows developers to obtain revenue through in-app advertising. Our study aims at identifying potential security risks of a type of mobile advertisement where advertisers are charged for their advertisements only when a user clicks (or touches) on the advertisements in their applications. In the Android platform, we design an automated click generation attack and empirically evaluate eight popular advertising networks by performing real attacks on them. Our experimental results show that six advertising networks (75%) out of eight (Millennial Media, App Lovin, Ad Fit, Mdot M, Rev Mob and Cauly Ads) are vulnerable to our attacks. We also discuss how to develop effective defense mechanisms to mitigate such automated click fraud attacks.

2.2 MAdFraud: Investigating ad fraud in Android applications:-

Many Android applications are distributed for free but are supported by advertisements. Ad libraries embedded in the app fetch content from the ad provider and display it on the app's user interface. The ad provider pays the developer for the ads displayed to the user and ads clicked by the user. A major threat to this ecosystem is ad fraud, where a miscreant's code fetches ads without displaying them to the user or "clicks" on ads automatically. Ad fraud has been extensively studied in the context of web advertising but has gone largely unstudied in the context of mobile advertising. We take the first step to study mobile ad-fraud perpetrated by Android apps. We identify two fraudulent ad behaviors in apps requesting ads while the app is in the background, and clicking on ads without user interaction. Based on these observations, we developed an analysis tool, MAdFraud, which automatically runs many apps simultaneously in emulators to trigger and expose ad fraud. Since the formats of ad impressions and clicks vary widely between different ad providers, we develop a novel approach for automatically identifying ad impressions and clicks in three steps: building HTTP request trees, identifying ad request pages using machine learning, and detecting clicks in HTTP request trees using heuristics.

2.3 Detecting click fraud in pay-per-click streams of online advertising networks:-

With the rapid growth of the Internet, online advertisement plays a more and more important role in the advertising market. One of the current and widely used revenue models for online advertising involves charging for each click based on the popularity of keywords and the number of competing advertisers. This pay-per-click model leaves room for individuals or rival companies to generate false clicks (i.e., click fraud), which pose serious problems to the development of healthy online advertising market. To detect click fraud, an important issue is to detect duplicate clicks over decaying window models, such as jumping windows and sliding windows. Decaying window models can be very helpful in defining and determining click fraud. However, although there are available algorithms to detect duplicates, there is still a lack of practical and effective solutions to detect click fraud in pay-per-click streams over decaying window models.

2.4 On hit inflation techniques and detection in streams of web advertising networks:-

Click fraud is jeopardizing the industry of Internet advertising. Internet advertising is crucial for the thriving of the entire Internet. Since it allows producers to advertise their products, and hence contributes to the well being of e-commerce. Moreover, advertising supports the intellectual value of the Internet by covering the running expenses of the content publishers' sites. Some publishers are dishonest, and use automation to generate traffic to defraud the advertisers. Similarly, some advertisers automate clicks on the advertisements of their competitors to deplete their competitors ' advertising budgets. We provide a complete classification of the hit inflation techniques; and devise stream analysis techniques that detect a variety of fraud attacks. We abstract detecting the fraud attacks of some classes as theoretical stream analysis problems that we bring to the data management research community as open problems.

CHAPTER 3

SYSTEM ANALYSIS

3.1 EXISTING SYSTEM

Some researchers have studied vulnerability detection and assessment. Some studies on vulnerability diagnosis, for example, Emeka and Liu proposed a method to identify software security vulnerabilities. However, their work mostly concerned software rather than systems themselves. Xu and Nygard, Fang *et al*, proposed a method to locate change based on behavioral profiles. They determined the different parts by comparing the differences between two models. However, this method was based on two models, and it was not suitable for a single model. In fact, in most cases, we do not know the meta-model. In our previous work we proposed a method to diagnose the vulnerability of a model. At the same time, we provided a method to locate the vulnerable points. However, this method could not evaluate these vulnerable points.

Allodi and Massacci considered two-stage attacks and escalated the attacks. The escalated attack could be performed by exploiting local vulnerabilities in the target. Wang *et al*, proposed a method for fault diagnosis of a timed Petri net (TPN). They used a fault diagnosis graph to diagnose an observable TPN. Li and Hadjicostis, proposed a method for estimating the minimum initial marking in labeled Petri nets. They used this method to determine the minimum number of resources. Prakash *et al*, proposed a method to perform online fault detection and isolate multiple faults. They analyzed global and local faults by using the global fault sensitivity signature matrix (GFSSM) and fault sensitivity signature matrix (FSSM). Lefebvre proposed a diagnosis decision method by analyzing observation sequences. However, this method was suitable for observable actions.

Some studies on vulnerability diagnosis, for example, Al-Dwairi and Kamala used security, privacy, design, and content to evaluate the quality of B2C e-commerce websites. They gave an evaluation pattern for B2C e-commerce. However, for an arbitrary e-commerce system, they did not give a clear determination and evaluation method. Wang *et al*, proposed a vulnerability evaluation method based on the attack graph. This method could address low complexity attack paths. However, it could not address attack paths with unobservable activities. Fonseca *et al*, gave a method and a

tool to evaluate security mechanisms. This method could determine the possibility of injecting realistic vulnerabilities in a website. However, this diagnostic method was suitable for the process model itself. Pedroni analyzed uncertainty modeling and quantification methods to assess reliability and risk. The analysis compared advanced methods for the modeling, simulation and analysis of safety-critical systems and infrastructure under uncertainty. Khalid *et al*, examined the degree of customer satisfaction with an e-commerce system. They used a modified American customer satisfaction index (ACSI) model to describe 149 online data points. They showed that customer expectations and e-commerce service quality could affect perceived value. Najafi introduced e-trust building models and provided a method to enhance e-commerce security.

Greener-Brzezinski *et al*, proposed a method based on spatial positioning, navigation, and timing (PNT) to analyze accuracy, continuity, and reliability. Cabasino *et al*, used Petri nets based on the notions of minimal explanations and markings to diagnose DESs. They assumed that fault events could be modeled by observable transitions. This method was suitable for observable transitions. This method had the computational advantage than the diagnose-based ones, but it was not suitable for unobservable transitions. Basile *et al*, proposed a state estimation and fault diagnosis method for a labeled-time Petri net. They used a modified state class graph (MSCG) to perform fault diagnosis. Shoukry *et al*. proposed a method to evaluate security. They used satisfiability modulo theory to address the complexity of secure state estimation. Bonhomme a method to assess the marking of an unlabeled P-time Petri net with unobservable transitions. This method used the candidate firing sequences to estimate the marking. Koga *et al*, used the full-state and associated output feedback control law to control and estimate the one-phase Stefan problem. Hu *et al*. proposed a method to determine the optimal marked signal distribution. They used a given distortion constraint and expected embedding rate to obtain the optimal distribution.

Drawbacks in Existing System:-

- There is no proper VULNERABILITY ANALYSIS AND EVALUATION method which is not in an existing system.
- There is no technique called hidden Markov model (HMM) to find vulnerability on datasets.

3.2 PROPOSED SYSTEM:

The purpose of vulnerability assessment is to find and control vulnerability. In fact, if we can control these unobservable actions and assess the occurrence probability of each path to the final condition, we can diagnose and assess the e-commerce system. Therefore, we emphasize unobservable actions.

Our goal is to determine how to use a formal method to identify the vulnerable actions of a VET-net system and to give a formal calculation method for the vulnerability degree of a VET-net. The contributions of this paper include:

- We propose the concept of vulnerable transitions, which includes not only vulnerable actions but also unobservable transitions. We present a new method to locate the vulnerable transitions of a VET-net.
- We use hidden Markov models (HMMs) to approximate the state reachability graph of a VET-net. Based on this, we give a vulnerable state evaluation method for a VET-net.

In the proposed system, the system shows the framework of our methods. In the first stage, by capturing the user transaction behaviors of the e-commerce trading process, we analyze the observable and unobservable actions and then construct a VET-net model.

Based on the VET-net, we can find all paths to the final condition then, according to the characteristics of behaviors, we can locate the vulnerable actions of VET-nets, locating the source of fragility of the trading system for the user in the second step. In the third step, we can obtain a hidden Markov model according to the reachability graph of the VET-net. According to the hidden Markov model and reachability graph, we can compute the vulnerability degree of the VET-net.

Advantages:-

- Customers can easily manage the e-coupons via their mobile devices or PCs.
- For easy management, most e-coupon services manage e-coupon information in a decentralized system.
- The information can be easily manipulated by an administrator.

CHAPTER 4

SYSTEM SPECIFICATIONS

4.1 SYSTEM REQUIREMENTS

4.1.1 Hardware System Requirements

- Processor - Pentium –IV
- RAM - 4 GB (min)
- Hard Disk - 20 GB
- Key Board - Standard Windows Keyboard
- Mouse - Two or Three Button Mouse
- Monitor - SVGA

4.1.2 Software System Requirements

- Operating System - Windows XP
- Coding Language - Java/J2EE(JSP, Servlet)
- Front End - J2EE
- Back End - MySQL

CHAPTER 5

SYSTEM STUDY

5.1 FEASIBILITY STUDY

The feasibility of the project is analyzed in this phase and business proposal is put forth with a very general plan for the project and some cost estimates. During system analysis the feasibility study of the proposed system is to be carried out. This is to ensure that the proposed system is not a burden to the company. For feasibility analysis, some understanding of the major requirements for the system is essential.

Three key considerations involved in the feasibility analysis are,

- Economical Feasibility
- Technical Feasibility
- Social Feasibility

5.1.1 ECONOMICAL FEASIBILITY

This study is carried out to check the economic impact that the system will have on the organization. The amount of fund that the company can pour into the research and development of the system is limited. The expenditures must be justified. Thus the developed system as well within the budget and this was achieved because most of the technologies used are freely available. Only the customized products had to be purchased.

5.1.2 TECHNICAL FEASIBILITY

This study is carried out to check the technical feasibility, that is, the technical requirements of the system. Any system developed must not have a high demand on the available technical resources. This will lead to high demands on the available technical resources. This will lead to high demands being placed on the client. The developed system must have a modest requirement, as only minimal or null changes are required for implementing this system.

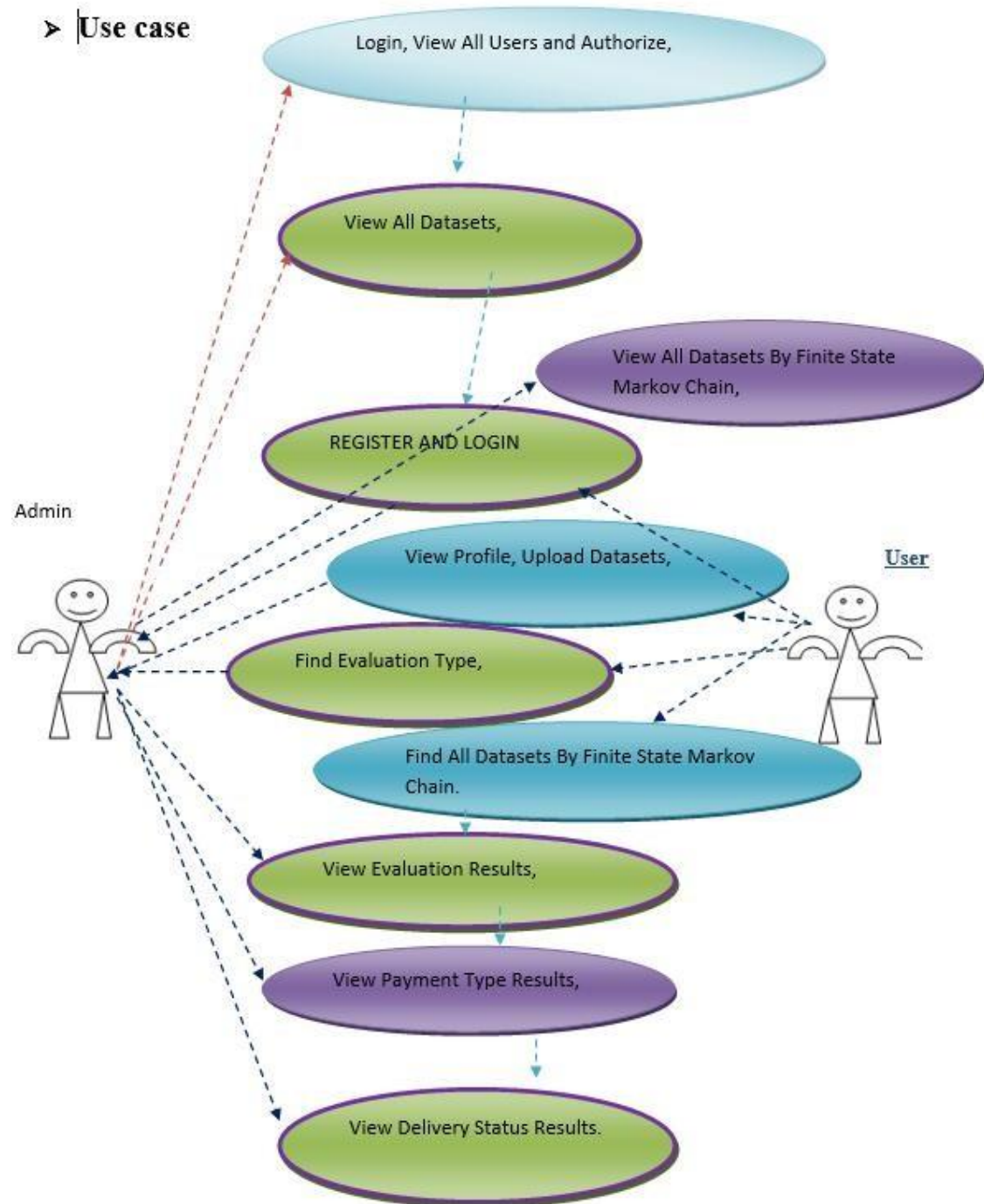
5.1.3 SOCIAL FEASIBILITY

The aspect of study is to check the level of acceptance of the system by the user. This includes the process of training the user to use the system efficiently. The user must not feel threatened by the system, instead must accept it as a necessity. The level of acceptance by the users solely depends on the methods that are employed to educate the user about the system and to make him familiar with it. His level of confidence must be raised so that he is also able to make some constructive criticism, which is welcomed, as he is the final user of the system.

CHAPTER 6

SYSTEM DESIGN

6.1. Use case diagram



6.2. Sequence Diagram

A sequence diagram in Unified Modeling Language (UML) is a kind of interaction diagram that show processes operate with one another and in what order. It is a construct of a Message Sequence Chart. Sequence diagrams are sometimes called event diagrams, event scenarios, and timing diagrams.

➤ Sequence Diagram

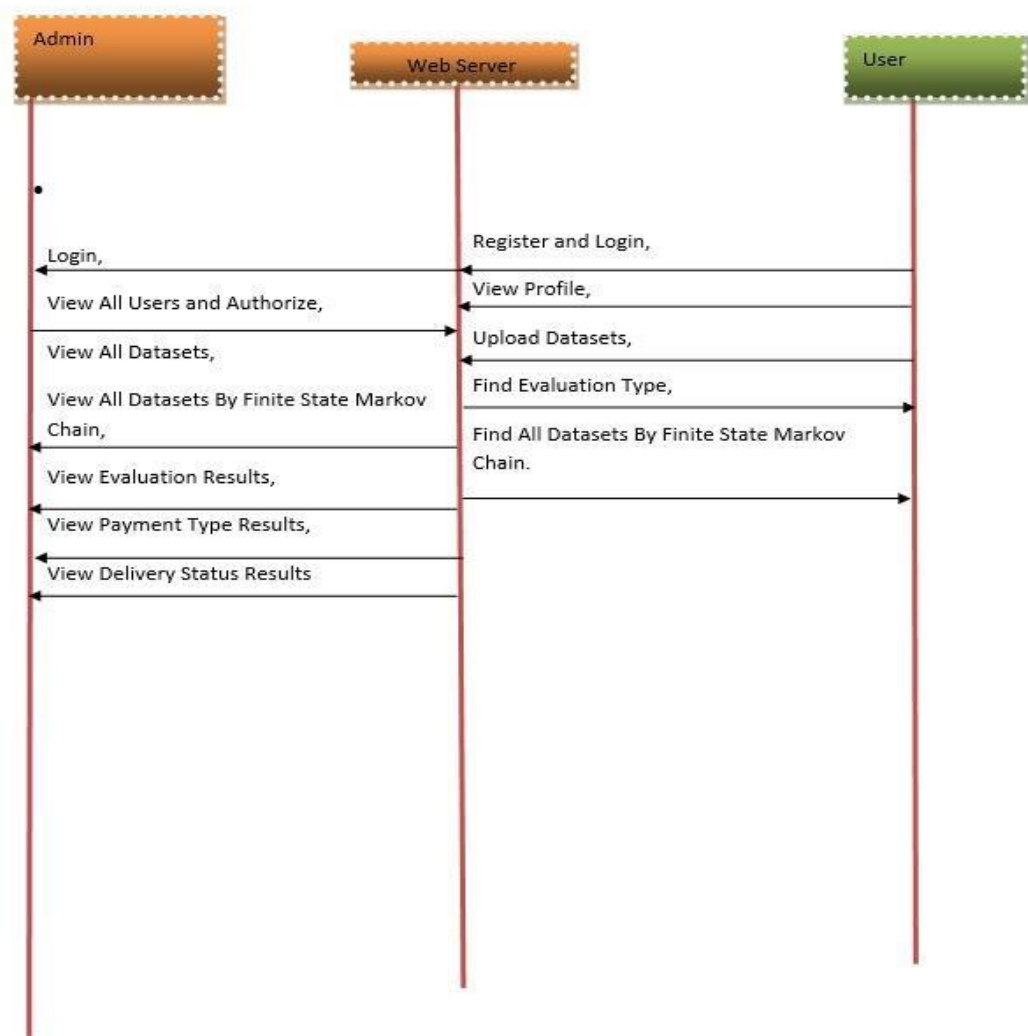
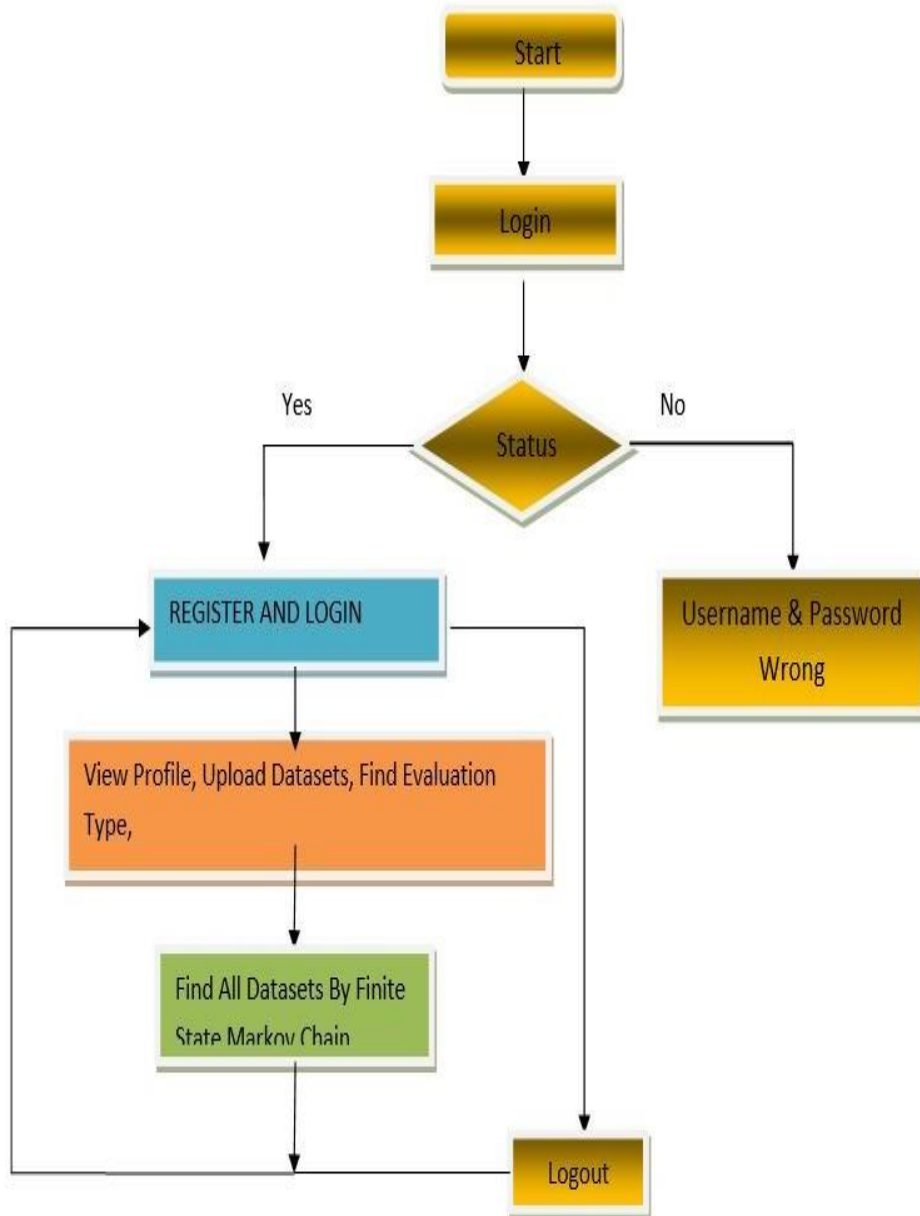


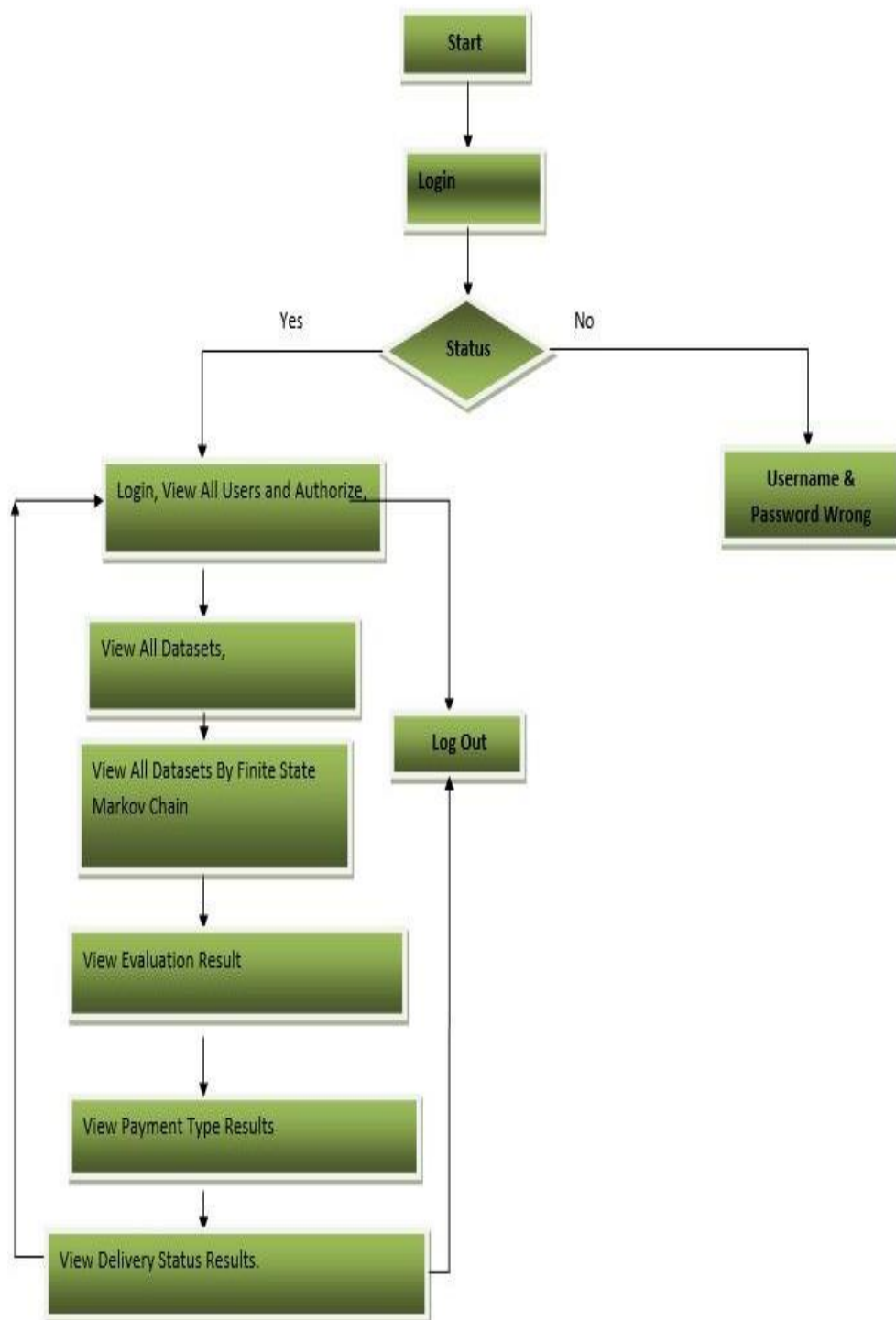
Fig.5. Sequence diagram

6.3. Flow Chart

➤ Flow Chart : User

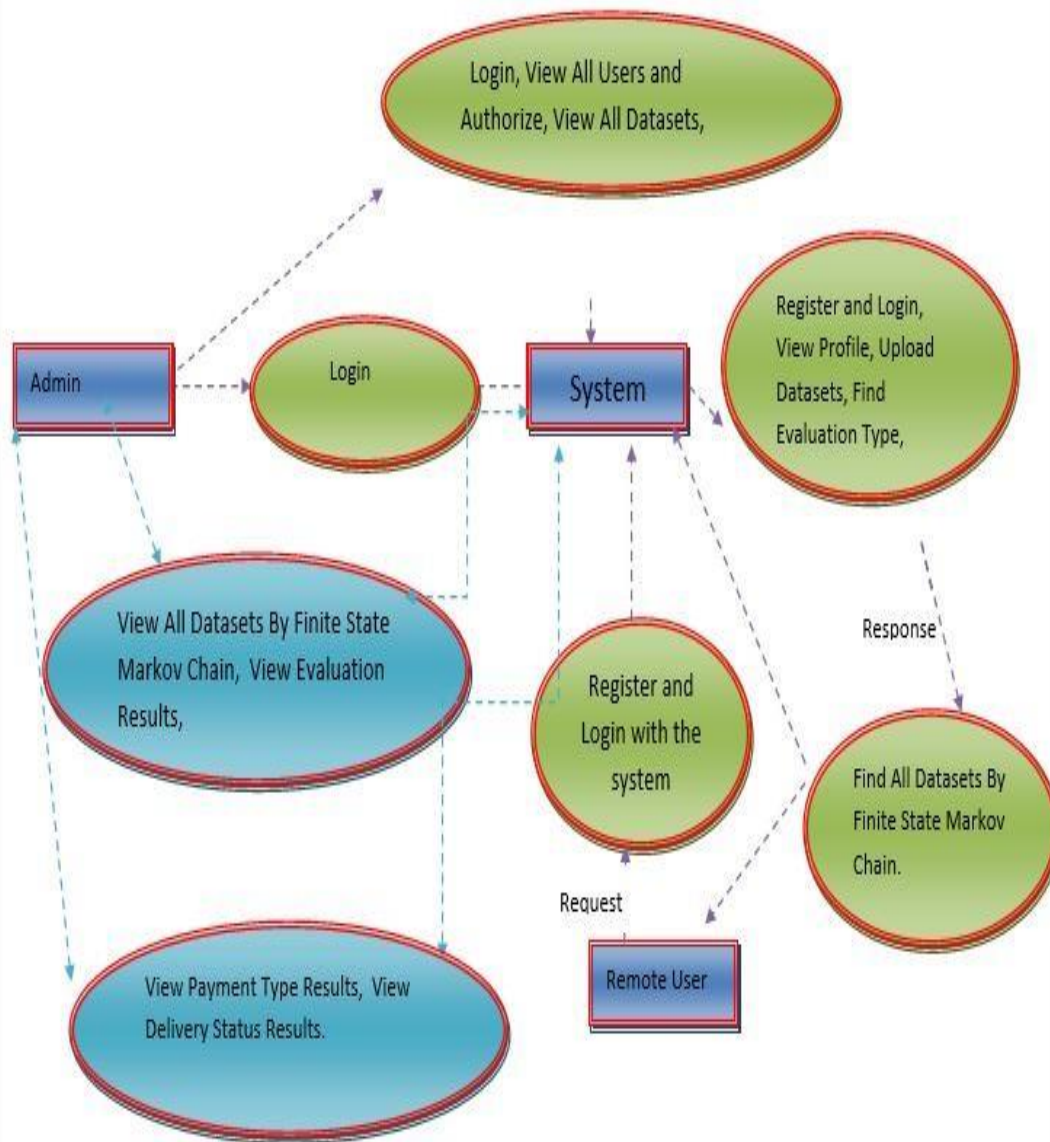


➤ Flow Chart : Admin



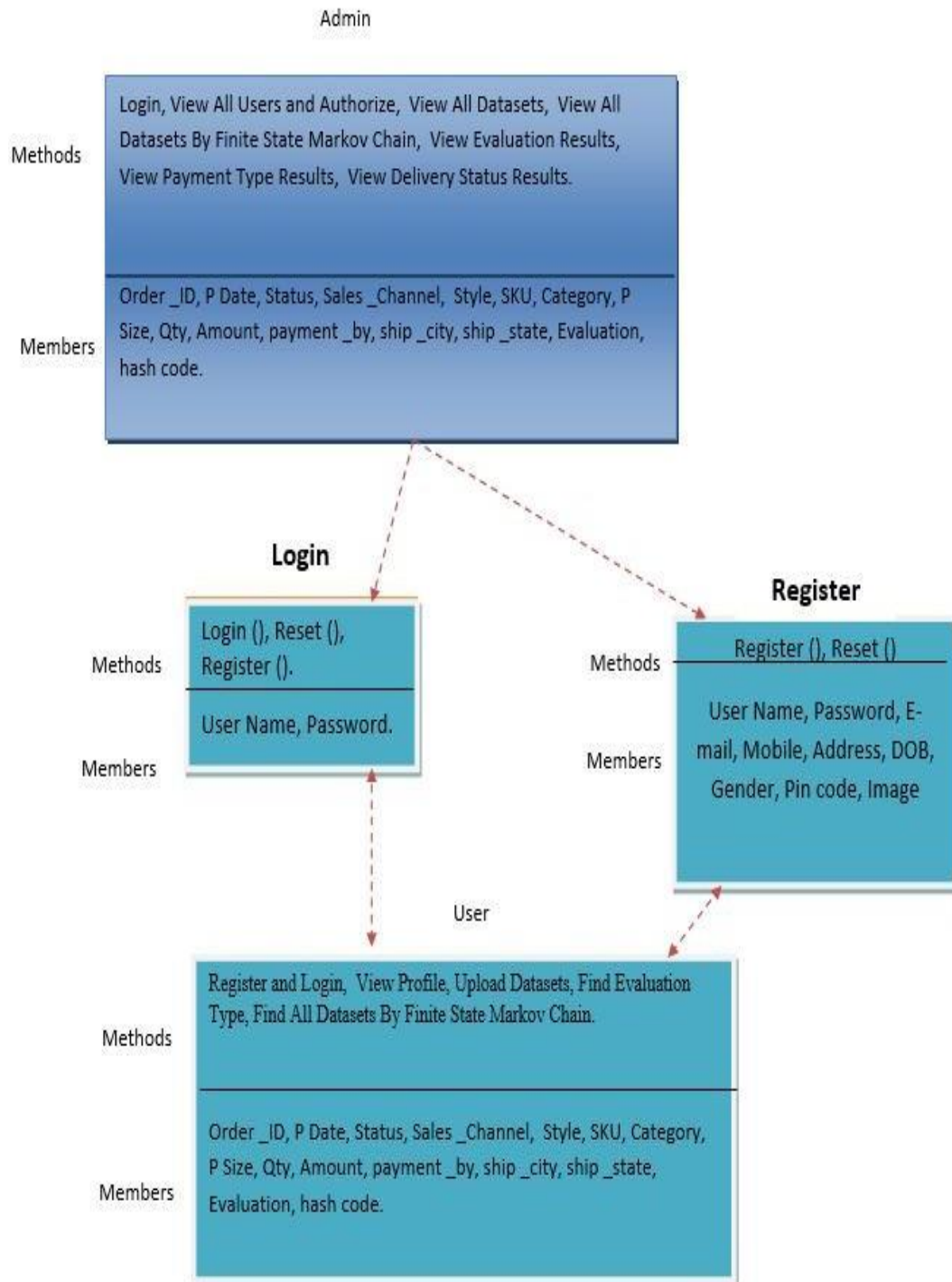
6.4. Data Flow Diagram

➤ Data Flow Diagram :



6.5. Class diagram

➤ Class Diagram :



CHAPTER 7

SYSTEM TESTING

The purpose of testing is to discover errors. Testing is the process of trying to discover every conceivable fault or weakness in a work product. It provides a way to check the functionality of components, sub assemblies, assemblies and/or a finished product. It is the process of exercising software with the intent of ensuring that the Software system meets its requirements and user expectations and does not fail in an unacceptable manner. There are various types of test. Each test type addresses a specific testing requirements.

7.1 Unit Testing

Unit testing involves the design of test cases that validate that the internal program logic is functioning properly, and that program inputs produce valid outputs. All decision branches and internal code flow should be validated. It is the testing of individual software units of the application. It is done after the completion of an individual unit before integration. This is a structural testing, that relies on knowledge of its construction and is invasive. Unit tests perform basic tests at component level and test a specific business process, application, and/or system configuration. Unit tests ensure that each unique path of a business process performs accurately to the documented specifications and contains clearly defined inputs and expected results.

Test objectives

- All field entries must work properly.
- Pages must be activated from the identified link.
- The entry screen, messages and responses must not be delayed.

Features to be tested

- Verify that the entries are of the correct format
- No duplicate entries should be allowed
- All links should take the user to the correct page.

7.2 Integration Testing

Integration tests are designed to test integrated software components to determine if they actually run as one program. Testing is event driven and is more

concerned with the basic outcome of screens or fields. Integration tests demonstrate that although the components were individually satisfactory, as shown by successful unit testing, the combination of components is correct and consistent. Integration testing is specifically aimed at exposing the problems that arise from the combination of components.

The following are the types of Integration Testing

7.2.1 Top Down Integration

This method is an incremental approach to the construction of program structure. Modules are integrated by moving downward through the control hierarchy, beginning with the main program module. The module subordinates to the main program module are incorporated into the structure in either a depth first or breadth first manner.

In this method, the software is tested from main module and individual stubs are replaced when the test proceeds downwards.

7.2.2 Bottom-up Integration

This method begins the construction and testing with the modules at the lowest level in the program structure. Since the modules are integrated from the bottom up, processing required for modules subordinate to a given level is always available and the need for stubs is eliminated. The bottom up integration strategy may be implemented with the following steps:

- The low-level modules are combined into clusters into clusters that perform a specific Software sub-function.
- A driver (i.e.) the control program for testing is written to coordinate test case input and output.
- The cluster is tested.
- Drivers are removed and clusters are combined moving upward in the program structure

The bottom up approaches tests each module individually and then each module is module is integrated with a main module and tested for functionality.

7.3. Functional Testing

Functional tests provide systematic demonstrations that functions tested are available as specified by the business and technical requirements, system documentation, and user manuals. Functional testing is centered on the following items:

Valid Input : identified classes of valid input must be accepted.

Invalid Input : identified classes of invalid input must be rejected.

Functions : identified functions must be exercised.

Output : identified classes of application outputs must be exercised.

Systems/Procedures: interfacing systems or procedures must be invoked.

Organization and preparation of functional tests is focused on requirements, key functions, or special test cases. In addition, systematic coverage pertaining to identify Business process flows; data fields, predefined processes, and successive processes must be considered for testing. Before functional testing is complete, additional tests are identified and the effective value of current tests is determined.

7.4. System Testing

System testing ensures that the entire integrated software system meets requirements. It tests a configuration to ensure known and predictable results. An example of system testing is the configuration oriented system integration test. System testing is based on process descriptions and flows, emphasizing pre-driven process links and integration points.

7.5. white box testing

White Box Testing is a testing in which the software tester has knowledge of the inner workings, structure and language of the software, or at least its purpose. It is used to test areas that cannot be reached from a black box level.

7.6. Black box testing

Black Box Testing is testing the software without any knowledge of the inner workings, structure or language of the module being tested. Black box tests, as most other kinds of tests, must be written from a definitive source document, such as

specification or requirements document, such as specification or requirements document. It is a testing in which the software under test is treated, as a black box .you cannot “see” into it. The test provides inputs and responds to outputs without considering how the software works.

7.7 Acceptance Testing

User Acceptance Testing is a critical phase of any project and requires significant participation by the end user. It also ensures that the system meets the functional requirements.

Test Results: All the test cases mentioned above passed successfully. No defects encountered.

CHAPTER 8

IMPLEMENTATION

8.1 Module Design

1.Issuer

In this module, the Admin has to login by using valid user name and password. After login successful he can do some operations such as Register and Login, View All Users And Authorize, View All Stores And Authorize, Add Category And Sub-Category, View All Products By Block chain, View All E Coupons By Block chain, View All E coupon Requested, View All Products Details, View All Users Search Transaction Categorized By Search Type, View All Users Search History, View All User Purchased Products, View All Keyword Facet, View Product's Rank In Chart.

2.Customer

In this module, there are n numbers of users are present. User should register with group option before doing some operations. After registration successful he has to wait for admin to authorize him and after admin authorized him. He can login by using authorized user name and password. Login successful he will do some operations like Register and Login, My Profile, Manage Accounts, Query Search By Keyword, All My Purchased Products Details, View All Search Transaction, View All My Search History, View Top K Searched Keyword Facets.

3.Store

In this module, there are n numbers of users are present. Transport Company user should register with group option before doing some operations. After registration successful he has to wait for admin to authorize him and after admin authorized him. He can login by using authorized user name and password. Login successful he will do some operations like Login, My Profile, Add Products, All My Product.

8.2 Architecture



CHAPTER 9

SOFTWARE ENVIRONMENT

9.1. Introduction to Java

The Java Language When it comes to Java, you're getting a language and a platform in one. Java, one of the most extensively used programming languages on the planet, is the programming language Java is described by the following buzzwords:

Object-oriented design

1. Conveniently sized
2. A Dispersed
3. Incredibly efficient
4. Translated
5. In a multi-threaded environment
6. Stable
7. Dynamic
8. Ensured

Compiling or interpreting a program written in one of the many popular programming languages allows you to run it on your computer. Java is an unusual programming language since it can be both compiled and interpreted. The compiler generates Java byte codes from platform-independent code, which is then interpreted by the Java platform's interpreter. The interpreter parses and executes each Java bytecode instruction on the computer. While compilation occurs only once, each time a program is run, interpretation occurs on every run. The graphic below illustrates this.

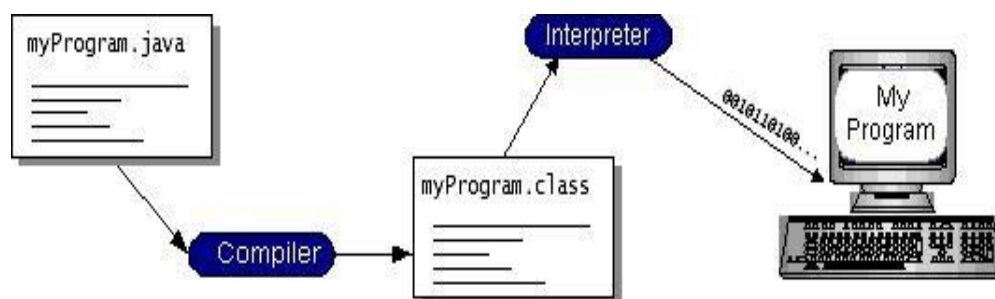


Fig.3. Process of Program execution

9.1.1. General J2ME architecture

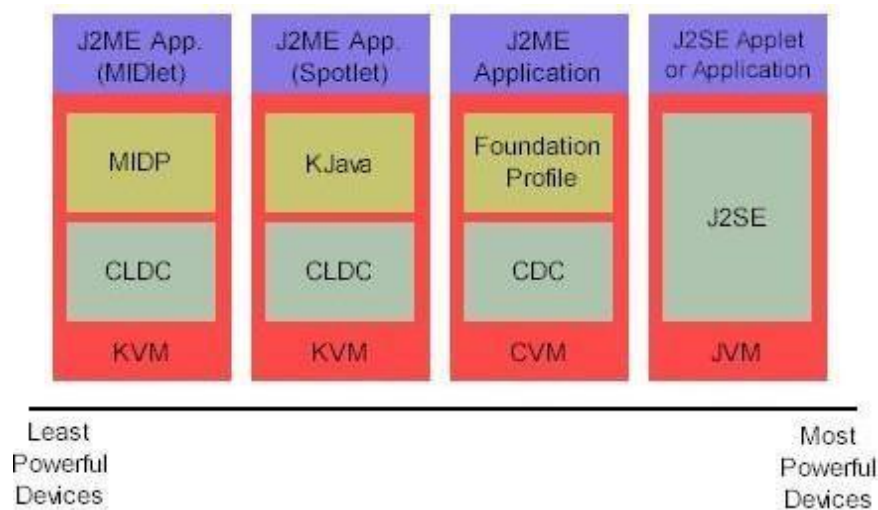


Fig.4. J2ME Architecture

Configurations and profiles are used by J2ME to customize the Java Runtime Environment (JRE). With the addition of domain-specific classes and a Java Runtime Environment (JRE), J2ME is a full Java Runtime Environment (JRE). This means that only certain types of hardware can run a specified set of core classes and a specific JVM. The course will go into great detail on how to set up various components. To put it another way, this profile adds specific domain-specific classes to the J2ME setup to identify specific uses for devices. The many virtual machines, settings, and profiles are depicted in the diagram below. We'll go into great detail on profiles. Java virtual machine and the J2SE API share many similarities. It is customary to refer to the Java Virtual Machine (JVM) as a whole when discussing the J2ME virtual machines (KVM and CVM). Although they are shortened versions of the J2SE JVM, the J2ME-specific KVM and CVM can be considered Java virtual machines. Building J2ME apps is the next stage.

Introduction creating apps for mobile devices of all sizes has a few considerations that will be explored in this section. If you're using J2SE to compile J2ME apps, look at how the compiler is called. The packaging and deployment procedure would be incomplete without pre verification.

In the design of compact devices, there are three factors to consider. When creating software for mobile devices, it's important to keep a few things in mind. It's a

good idea to go through your use case before diving into the code for a small device. It can be a frustrating experience to have to redo the code since you failed to account for all of the "gotchas". Take a look at the following options:

Don't complicate things. You can either remove unnecessary functionality or create a separate app for them.

It is advisable to have a smaller size. This should be taken into consideration by all developers. Because they are smaller, apps can be downloaded and installed more rapidly. When it comes to distributing your apps, think about using Java Archive (jar) files.

Keep RAM usage to a minimum while the app is running. The amount of memory required at run time can be reduced by using scalar rather than object types. Don't trust the trash hauler, either. Memory can be conserved by converting unused object references to null. Allowing objects to be allocated only when they are needed decreases the amount of memory required at runtime. In order to limit memory consumption on small devices, it is important to release resources as soon as possible, reuse objects, and avoid exceptions.

This means that only certain types of hardware can run a specified set of core classes and a specific JVM. There are currently just two J2ME configurations available:

In order to use the KVM's Connected Limited Device Configuration, you must use 16-bit or 32-bit devices with a small amount of memory (CLDC). This configuration is commonly used for developing small J2ME apps (along with the corresponding virtual machine). CLDC is more difficult to develop than CDC due to its smaller size. CLDC is the setting we'll be using for our drawing tool application. A small wireless device running a small application is an example of a Palm hand-held computer.

Configured Device Configuration (CVC) is used by the CVM on 32-bit systems requiring more than 2 MB of memory (CDC). Devices like the Net TV box fall under this category.

J2ME profiles number five In this session, we spoke about the importance of a profile in determining which devices are supported.

the Mobile Information Device Profile (MIDP) establishes the class of a mobile phone (MIDP). It adds domain-specific classes to the J2ME configuration in order to find usage for related devices. KJava and MIDP are the two CLDC-based J2ME profiles. Both KJava and MIDP are connected with CLDC and smaller devices. Profiles are built on configurations. Memory (memory capacity) of the device on which an application is running is linked to certain profile settings.

There is a Foundation Profile from which you can design your own, a skeleton profile.

KJava is the first profile in this list.

The KJava API is contained in a Sun profile named KJava, which is Sun's proprietary profile. The KJava profile is constructed from the CLDC setup. The KVM virtual machine accepts the same byte codes and class file formats as the J2SE virtual machine. KJava includes the Sun-specific API, which runs on Palm OS. Both the J2SE AWT and KJava API have a lot of similarities (AWT). So its default J2ME package has been replaced by `com.sun.kjava`. A better understanding of the KJava API will be gained after writing some sample programs.

In this section, you will find the MIDP profile:

MIDP is primarily aimed for mobile devices like smartphones and pagers. It is possible to dynamically install new applications and services on end-user devices thanks to the MIDP, which is based on CLDC, like KJava. Because it is not specific to any one manufacturer, MIDP has become a mobile device industry standard. It's an all-in-one development platform for mobile apps, complete with comprehensive documentation and assistance. It's worth noting that only the first three of the following packages are CLDC-specific.

The `java.lang`

package contains `IOException` in the JDK

Interfaces in the `java.util`

The Java Micro Edition Interface is implemented by this class (Interface for Micro edition Objects in Java (LCDUI))

* `javax.microedition.microedition`

CHAPTER 10

RESULTS

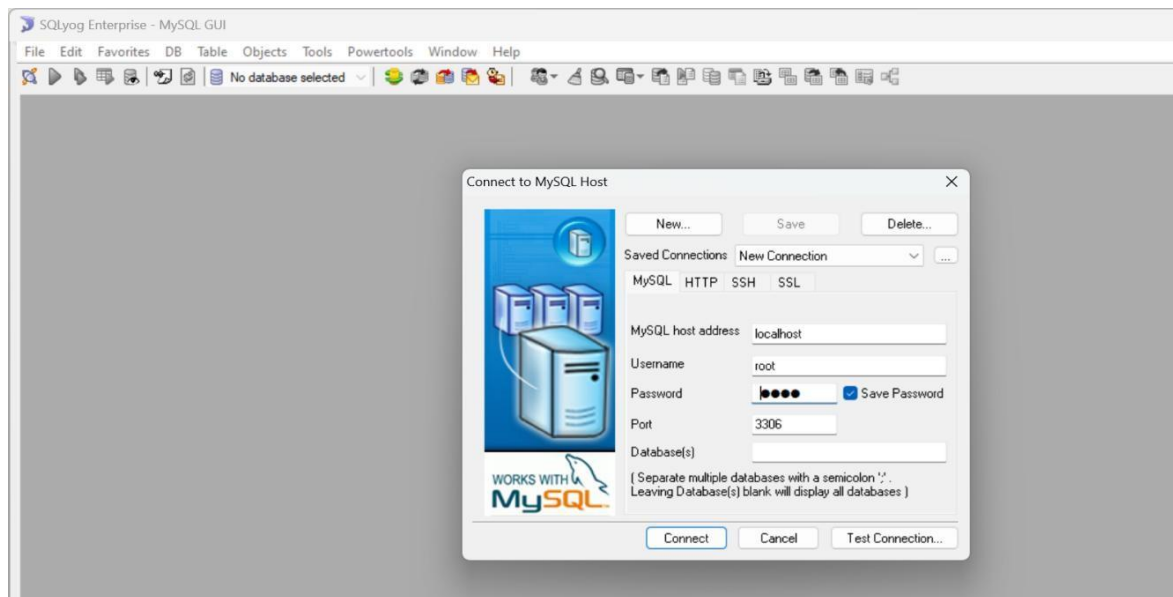


Fig : Connecting MYSQL

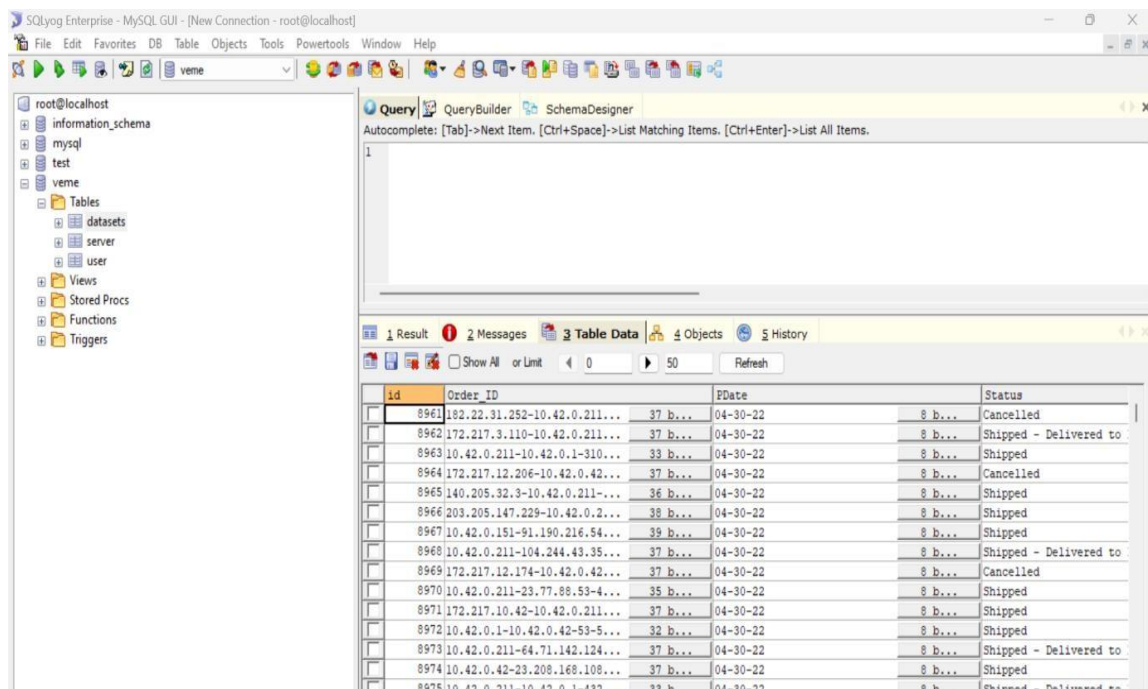


Fig : Storing Data Collection

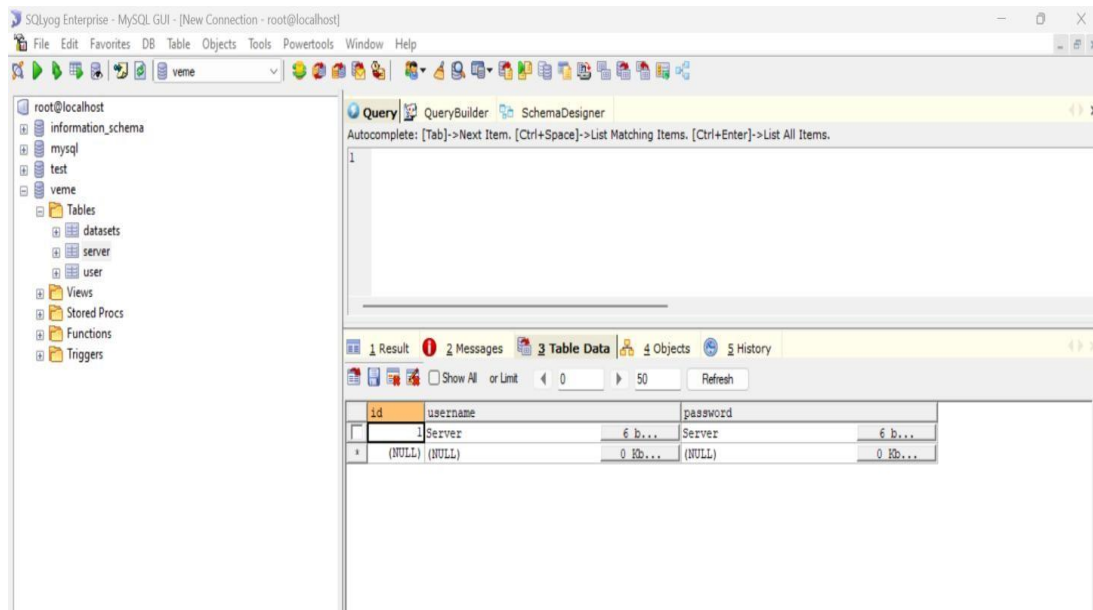


Fig : Admin Information

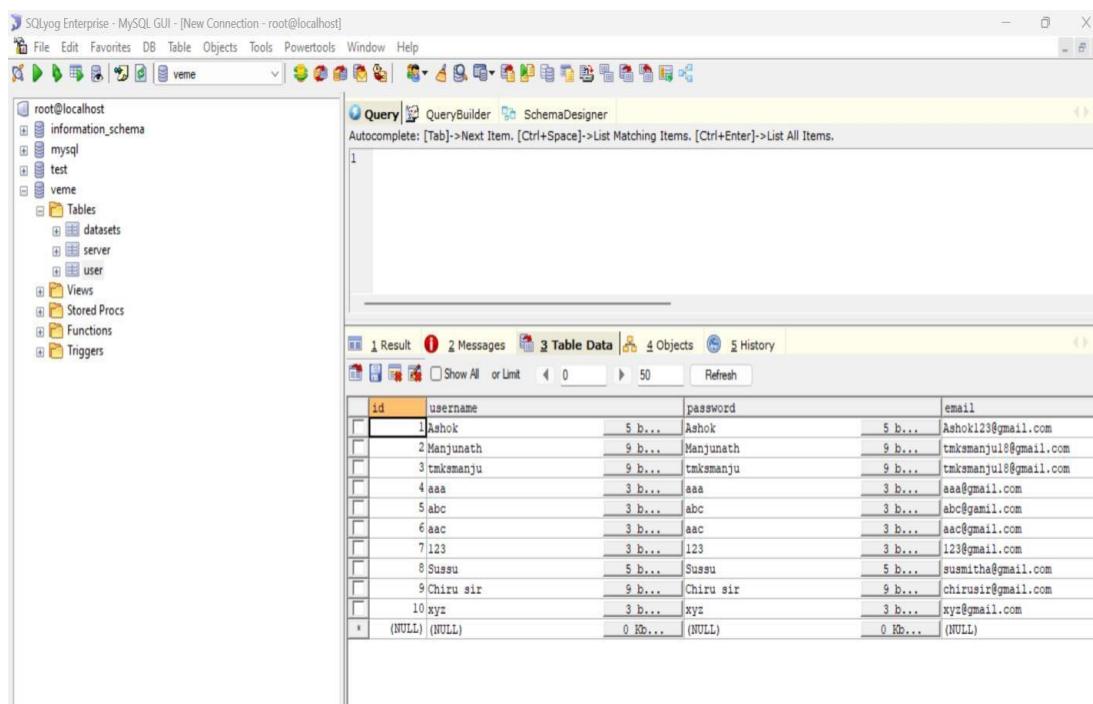


Fig : Storing Information

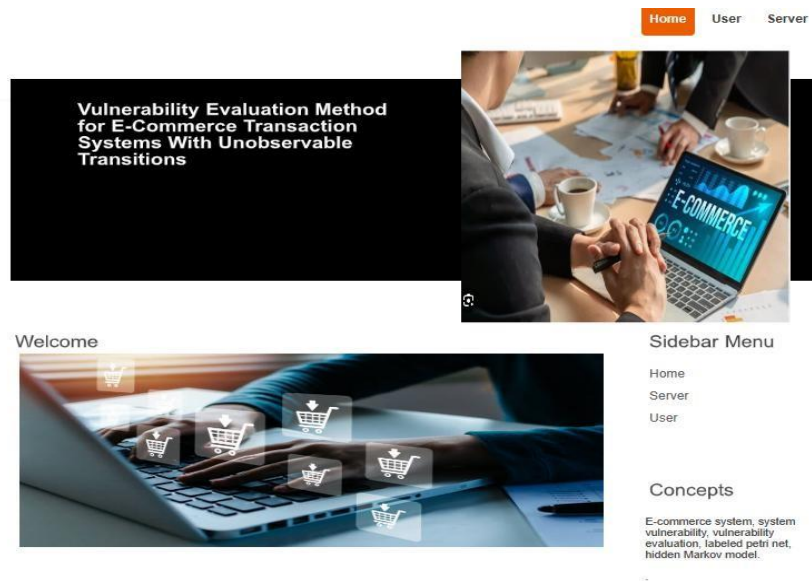


Fig : Home Page

Vulnerability Evaluation Method for E-Commerce Transaction Systems With Unobservable Transitions

Welcome to Registration Form

User Name (required)

Password (required)

Email Address (required)

Mobile Number (required)

Date of Birth (required)

Select Gender (required)

Address

Enter Pincode (required)

Select Network (required)

Select Profile Picture (required) No file chosen

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Sidebar Menu

- Home
- Server
- User

Fig : User Registration Page

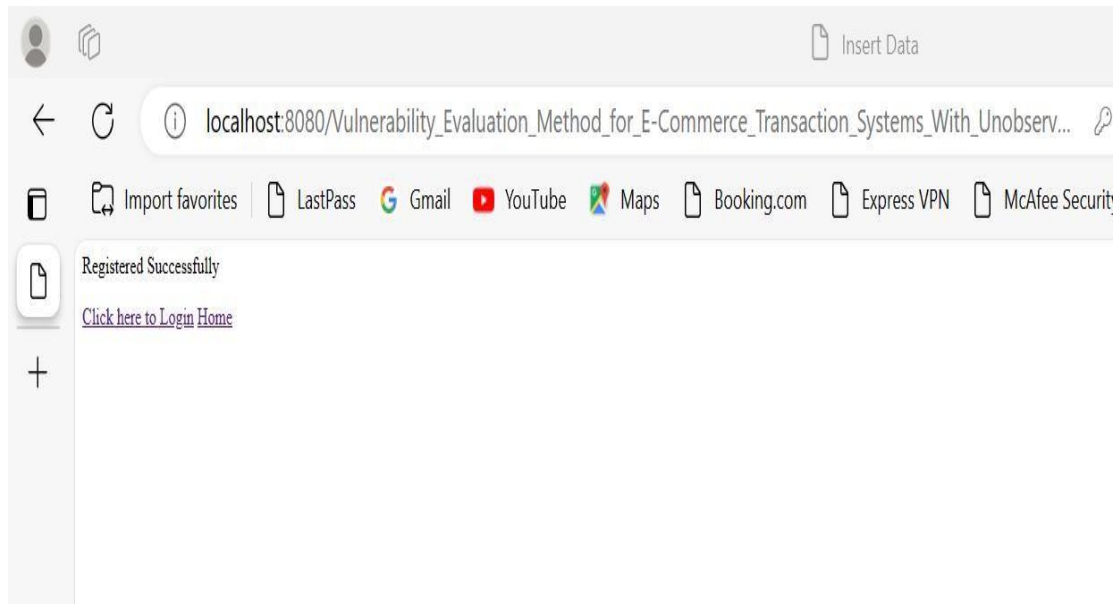


Fig : User Registration Successfull

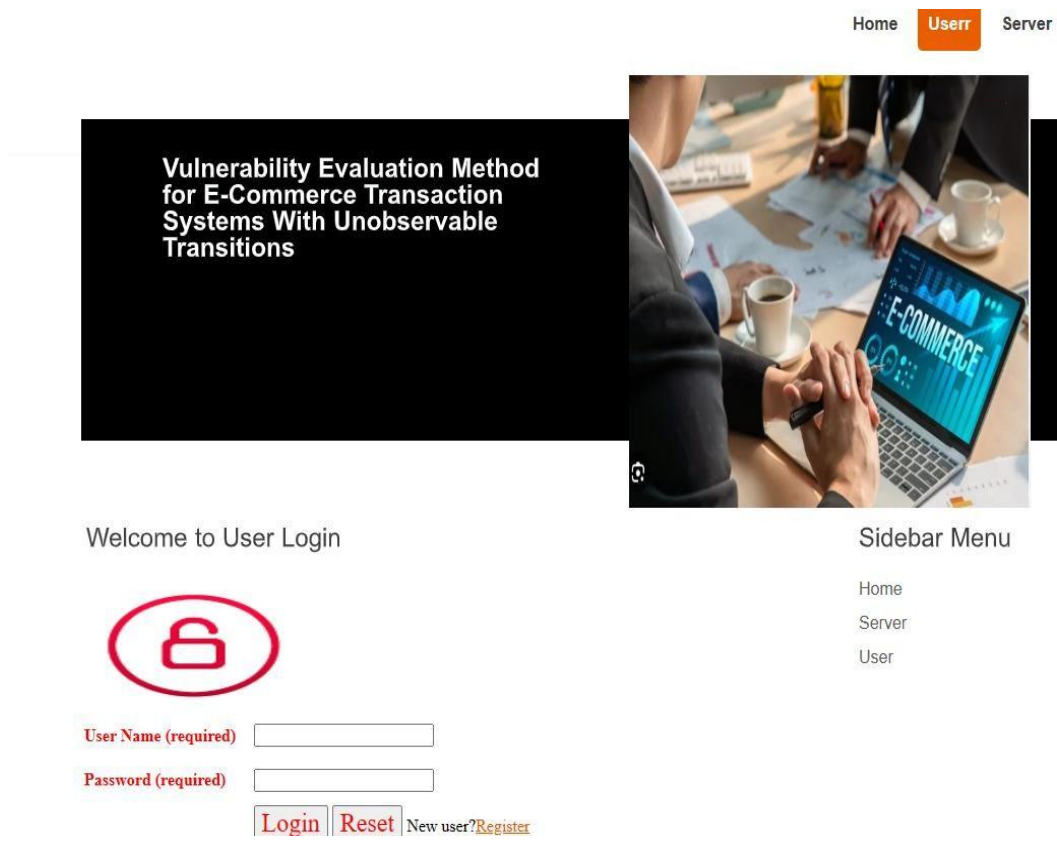


Fig : User Login Page

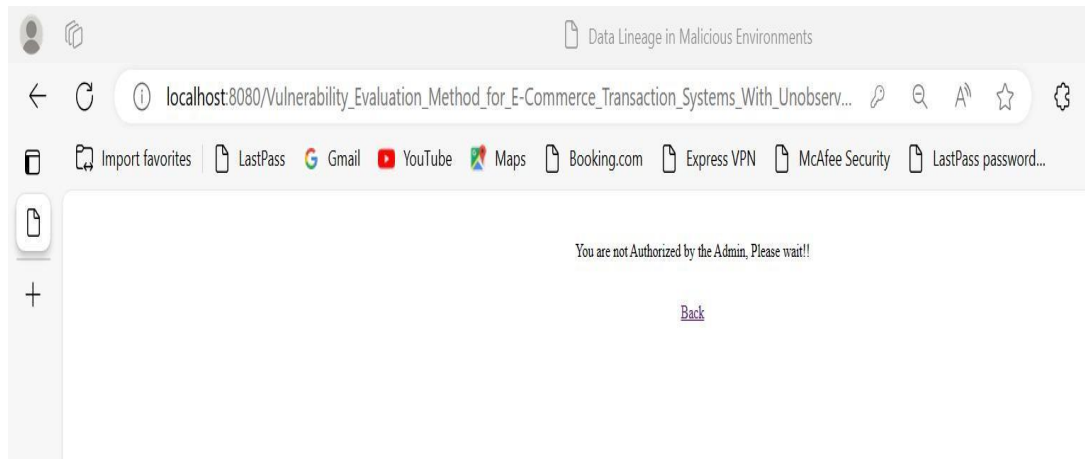


Fig : Authentication Page

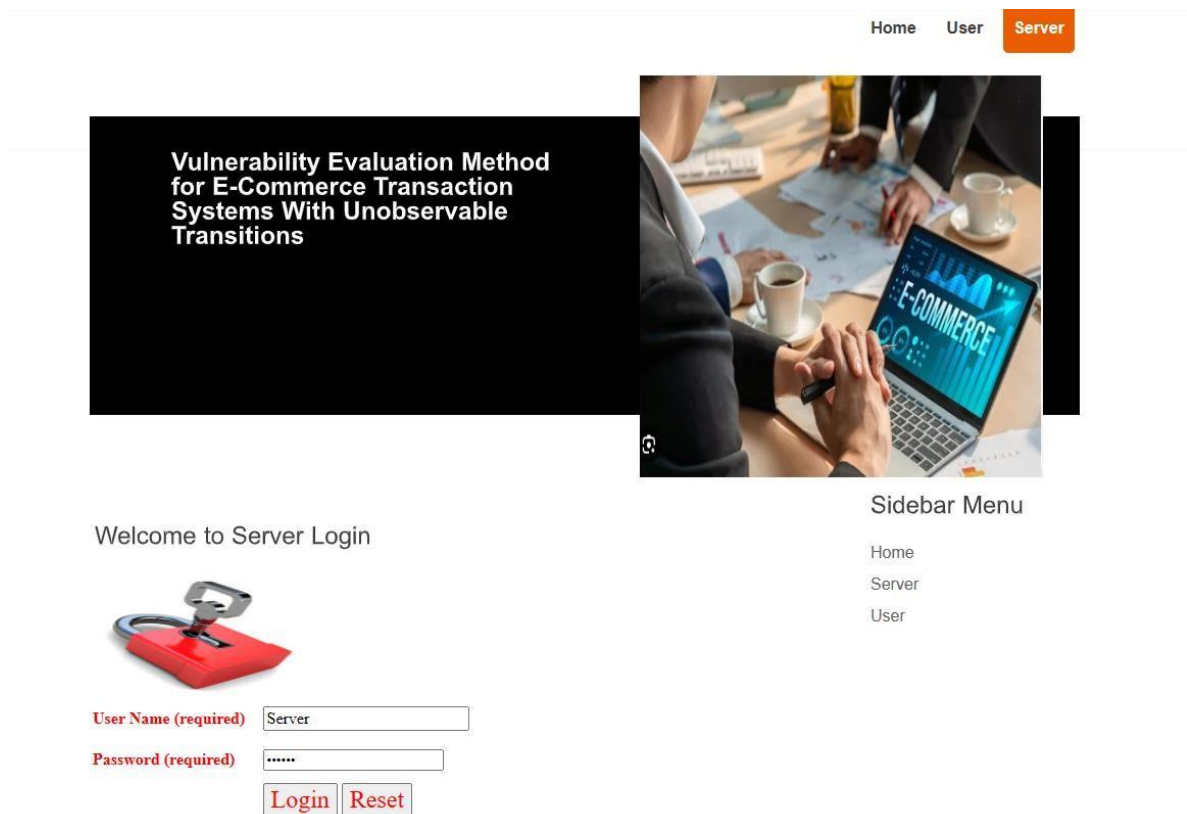


Fig : Server Login Page

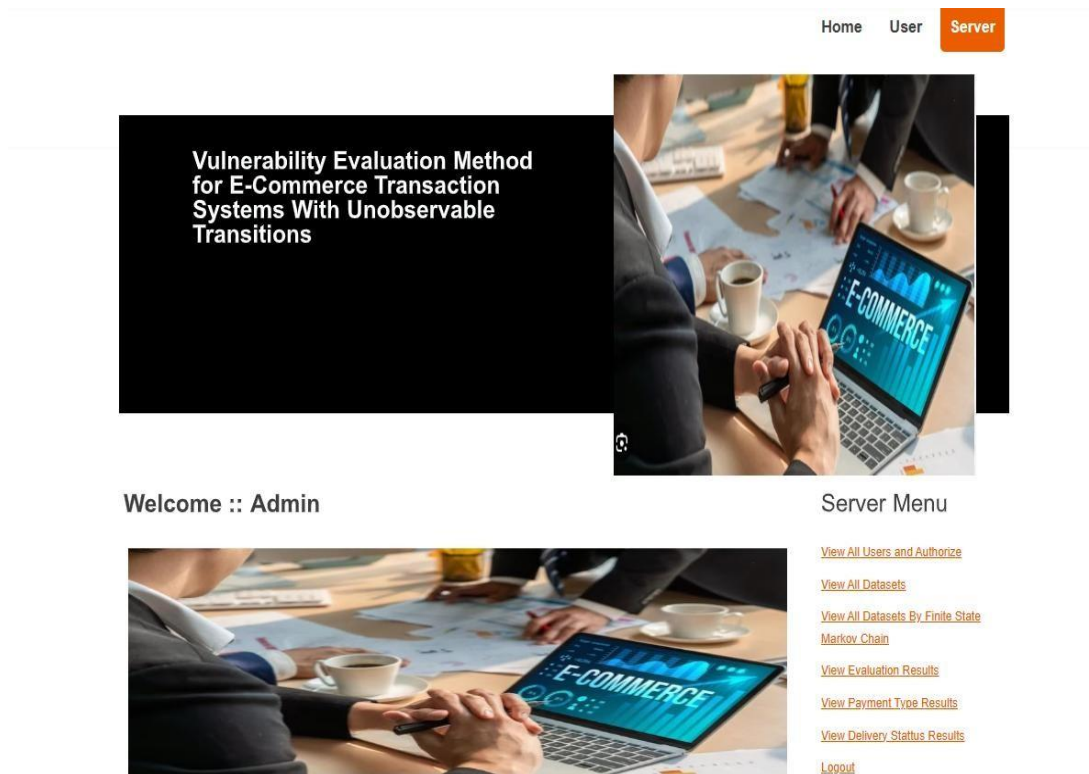


Fig : Admin Menu Page

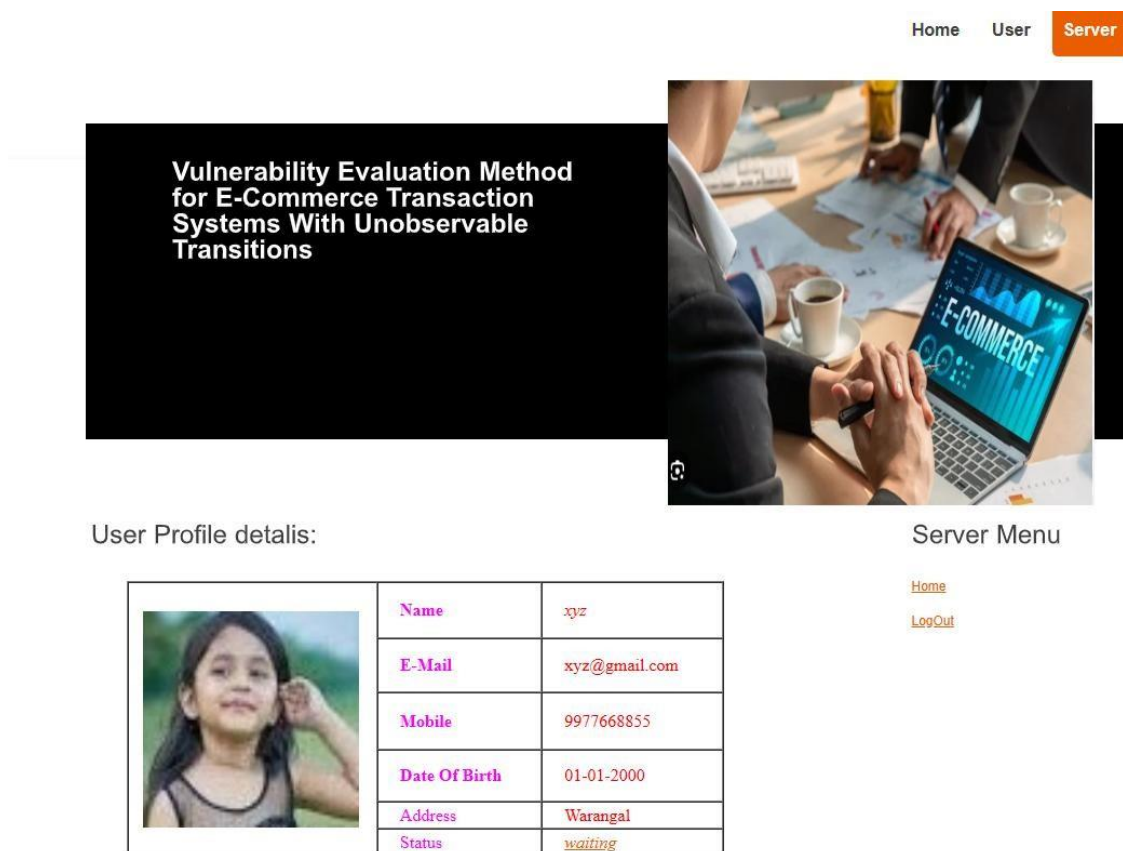


Fig : User Profile

[Home](#)
[User](#)
[Server](#)

Vulnerability Evaluation Method
for E-Commerce Transaction
Systems With Unobservable
Transitions



View All User Details...

Username	Email	mobile	View
Ashok	Ashok123@gmail.com	9535866270	more info...
Manjunath	tmksmanju18@gmail.com	9535866270	more info...
tmksmanju	tmksmanju18@gmail.com	9535866270	more info...
aaa	aaa@gmail.com	9999999999	more info...
abc	abc@gamil.com	9999999999	more info...
aac	aac@gmail.com	9999999999	more info...
123	123@gmail.com	9911919191	more info...
Sussu	susmitha@gmail.com	9988990066	more info...
Chiru sir	chirusir@gmail.com	9988776655	more info...
xyz	xyz@gmail.com	9977668855	more info...

Server Menu
[Home](#)
[LogOut](#)

Fig : All Users Details

[Home](#)
[User](#)
[Server](#)

Vulnerability Evaluation Method
for E-Commerce Transaction
Systems With Unobservable
Transitions



User Profile details:

	Name	xyz
	E-Mail	xyz@gmail.com
	Mobile	9977668855
	Date Of Birth	01-01-2000
	Address	Warangal
	Status	Authorized

Server Menu
[Home](#)
[LogOut](#)

Fig : User Profile Details

Vulnerability Evaluation Method for E-Commerce Transaction Systems With Unobservable Transitions



View All Datasets !!!

Order_ID	PDate	Status	Sales_Channel	Style	SKU	Category	PSize	Qty	Amount	payment_by
182.22.31.252-10.42.0.211-443-48485-6	04-30-22	Cancelled	Amazon.in	SET389	SET389-KR-NP-S	Set	S	0.0	647.62	Credit Card
172.217.3.110-10.42.0.211-443-60421-6	04-30-22	Shipped - Delivered to Buyer	Amazon.in	JNE3781	JNE3781-KR-XXXL	kurta	3XL	1.0	406.0	Credit Card
10.42.0.211-10.42.0.1-31065-53-17	04-30-22	Shipped	Amazon.in	JNE3371	JNE3371-KR-XL	kurta	XL	1.0	329.0	Credit Card
172.217.12.206-10.42.0.42-443-35767-6	04-30-22	Cancelled	Amazon.in	J0341	J0341-DR-L	Western Dress	L	0.0	753.33	Credit Card
140.205.32.3-10.42.0.211-443-55083-6	04-30-22	Shipped	Amazon.in	JNE3671	JNE3671-TU-XXXL	Top	3XL	1.0	574.0	Credit Card
203.205.147.229-10.42.0.211-80-52309-6	04-30-22	Shipped	Amazon.in	SET264	SET264-KR-NP-XL	Set	XL	1.0	824.0	Debit card
10.42.0.151-91.190.216.54-54248-12350-6	04-30-22	Shipped	Amazon.in	J0095	J0095-SET-L	Set	L	1.0	653.0	Credit Card
10.42.0.211-104.244.43.35-41821-443-6	04-30-22	Shipped - Delivered to Buyer	Amazon.in	JNE3405	JNE3405-KR-S	kurta	S	1.0	399.0	Credit Card

[Back](#)

Fig : Product Details

Vulnerability Evaluation Method for E-Commerce Transaction Systems With Unobservable Transitions



View All Datasets By Finite State Markov Chain !!!

Finite state Markov chain by Evaluation--->:: Vulnerability Found in ECommerce
Evaluation Type Mash Code --->::2a604c8ccea0b2a80ddef32d1742396b46e2cf

Order_ID	PDate	Status	Sales_Channel	Style	SKU	Category	PSize	Qty	Amount	payment_by
182.22.31.252-10.42.0.211-443-48485-6	04-30-22	Cancelled	Amazon.in	SET389	SET389-KR-NP-S	Set	S	0.0	647.62	Credit Card
172.217.3.110-10.42.0.211-443-60421-6	04-30-22	Shipped - Delivered to Buyer	Amazon.in	JNE3781	JNE3781-KR-XXXL	kurta	3XL	1.0	406.0	Credit Card
172.217.12.206-10.42.0.42-443-35767-6	04-30-22	Cancelled	Amazon.in	J0341	J0341-DR-L	Western Dress	L	0.0	753.33	Credit Card
10.42.0.151-91.190.216.54-54248-12350-6	04-30-22	Shipped	Amazon.in	J0095	J0095-SET-L	Set	L	1.0	653.0	Credit Card

Download

Fig : Finding Vulnerability

Vulnerability Evaluation Method for E-Commerce Transaction Systems With Unobservable Transitions



View Evaluation Results !!!...

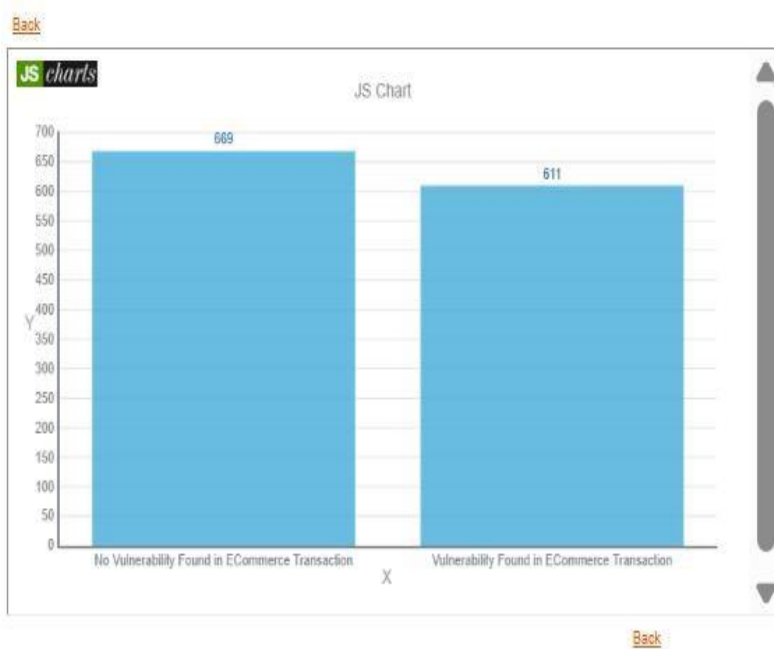


Fig : Finding Evaluation Results

Vulnerability Evaluation Method for E-Commerce Transaction Systems With Unobservable Transitions



View Payment Type Results !!!...

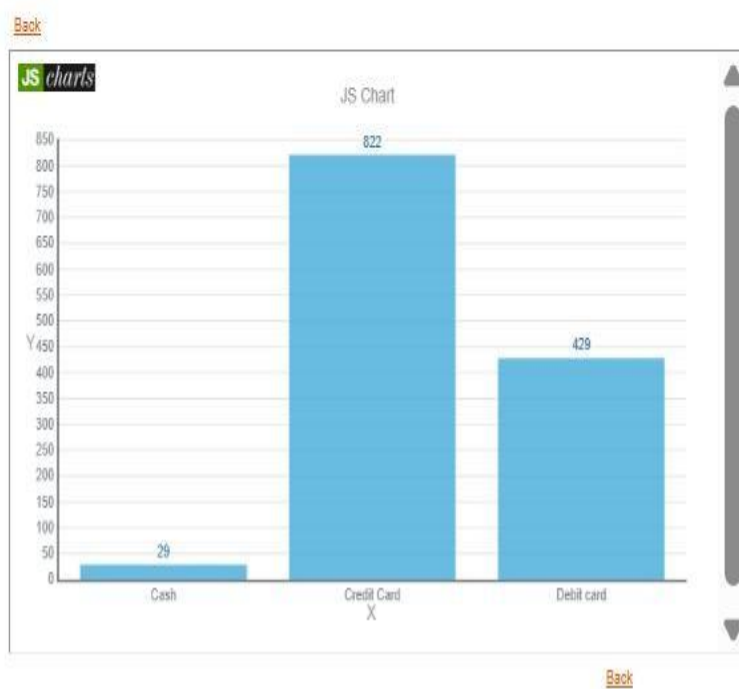


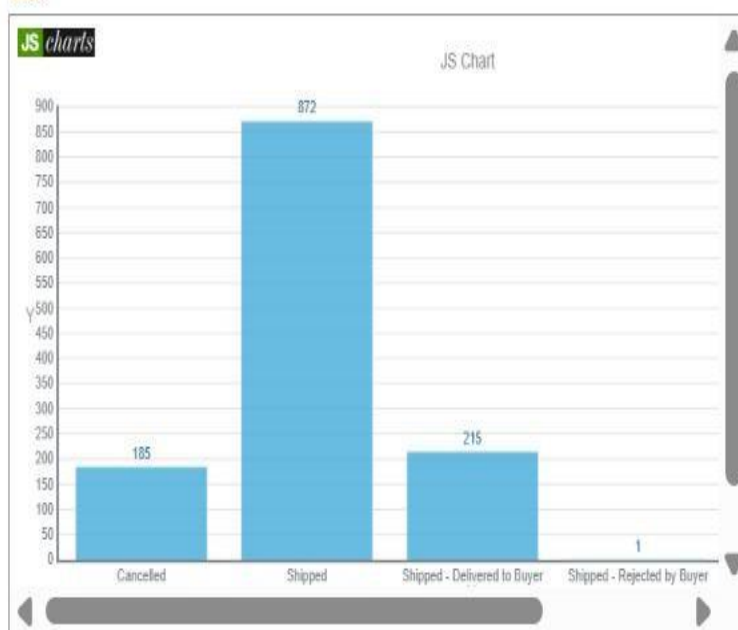
Fig : Payment Type Results

Vulnerability Evaluation Method for E-Commerce Transaction Systems With Unobservable Transitions



View Delivery Results !!!...

[Back](#)



[Back](#)

Fig : Viewing Delivery Details

Vulnerability Evaluation Method for E-Commerce Transaction Systems With Unobservable Transitions



Hi xyz , Your Profile details:

Sidebar Menu

	Name	xyz
	E-Mail	xyz@gmail.com
	Mobile	9977668855
	Date Of Birth	01-01-2000
	Address	Warangal

[Home](#)

[LogOut](#)

[Back](#)

Fig : User Profile Details

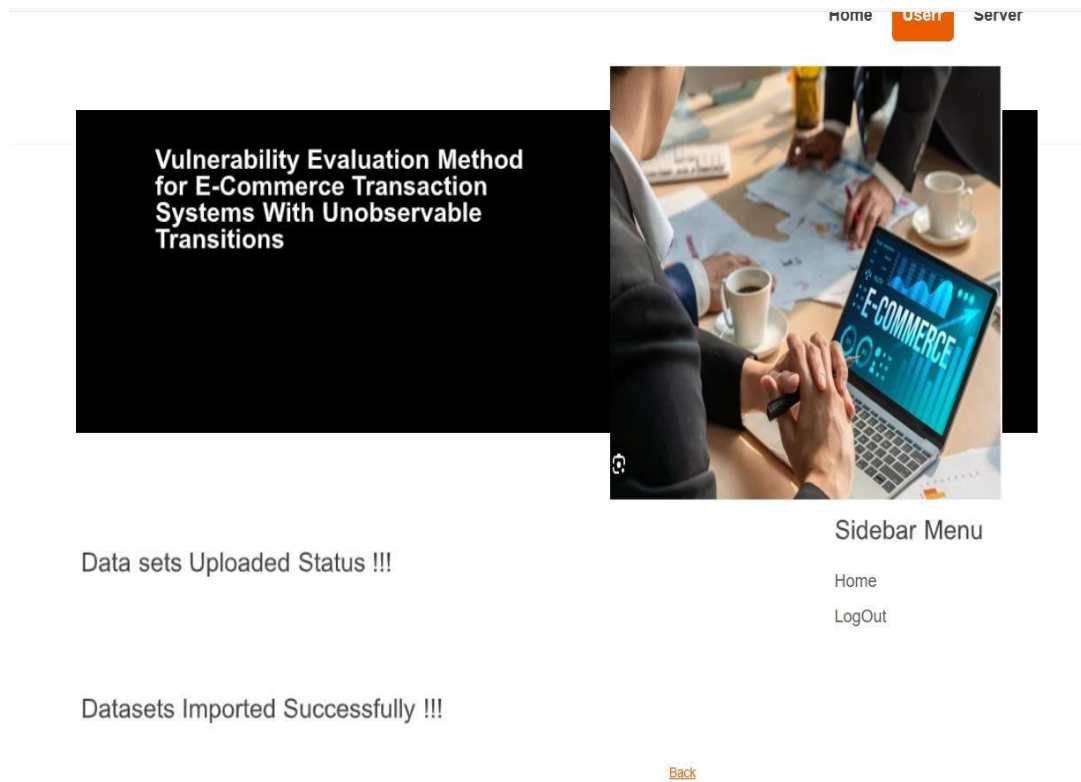


Fig : Importing Datasets

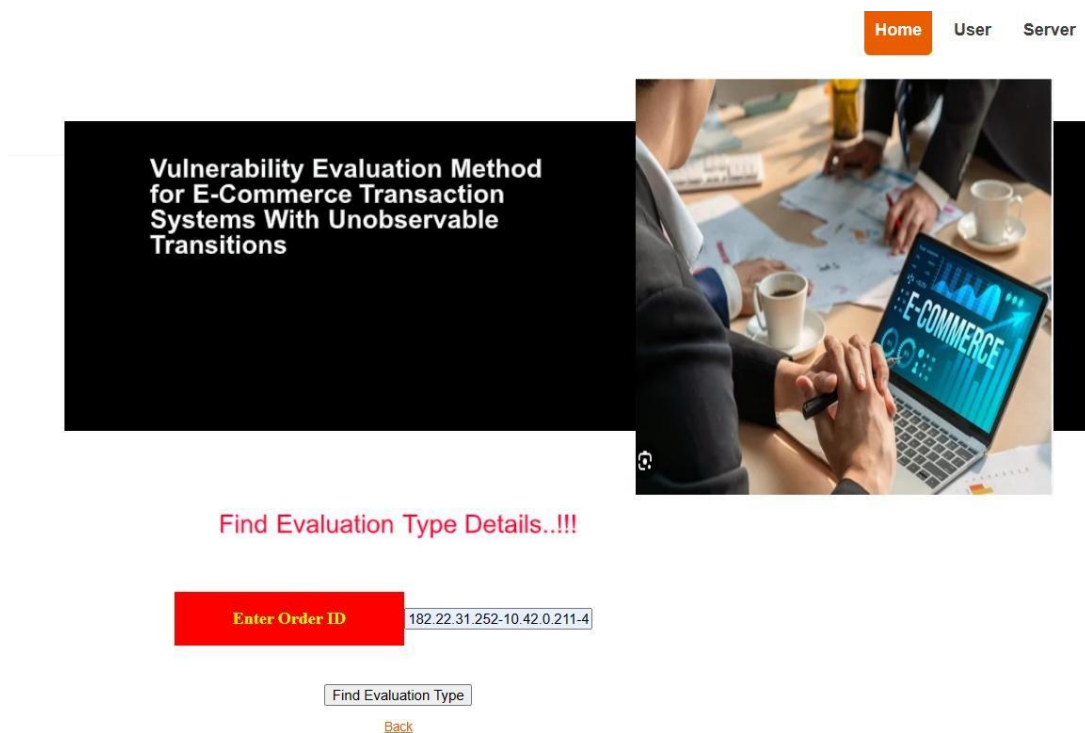


Fig : Finding Evaluation Type

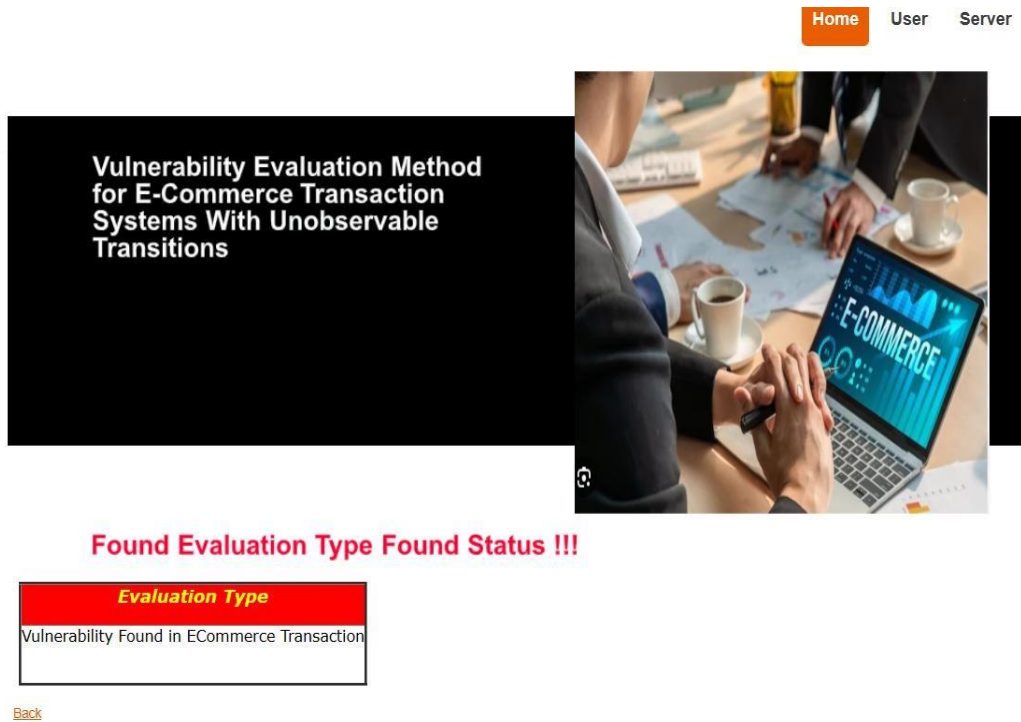


Fig : Evaluation Type status

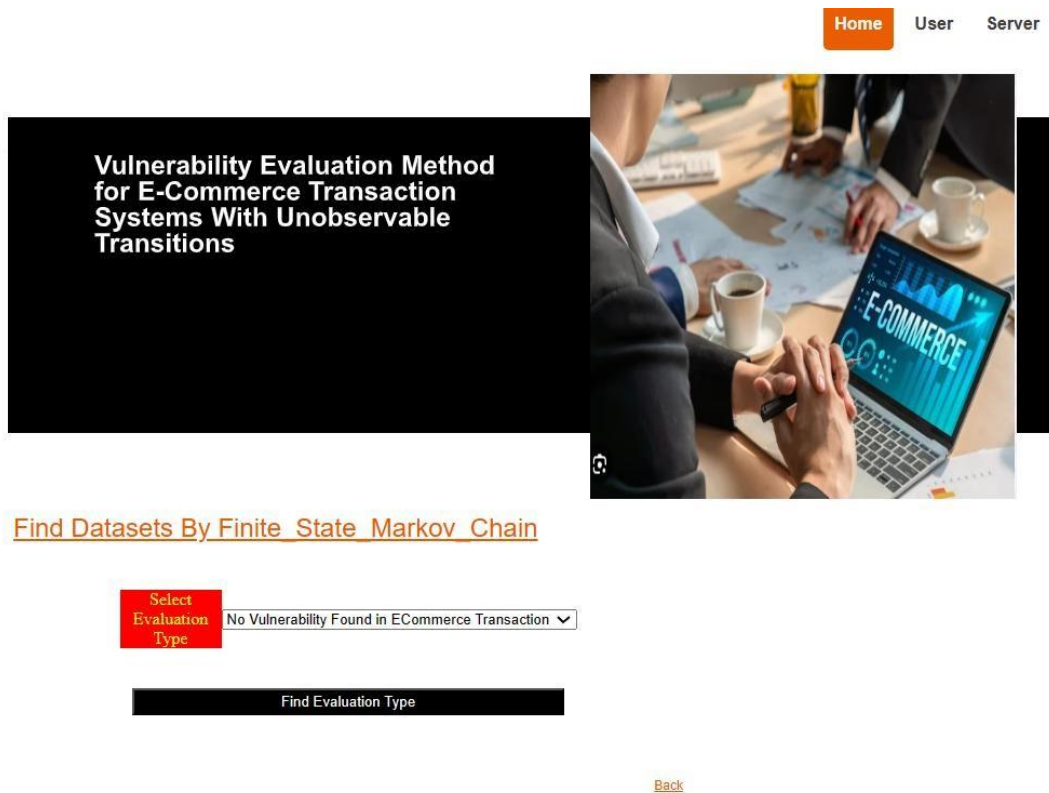


Fig : Finding Vulnerability Type

Home User Server

Vulnerability Evaluation Method
for E-Commerce Transaction
Systems With Unobservable
Transitions



Find Datasets By Finite_State_Markov_Chain

Order_ID	PDate	Status	Sales_Channel	Style	SKU	Category	PSize	Qty	Amount	payment_by	ship_city	ship_state	Evaluation	hashcode
10.42.0.211-10.42.0.1-31065-53-17	04-30-22	Shipped	Amazon.in	JNE3371	JNE3371-KR-XL	kurta	XL	1.0	329.0	Credit Card	NAVI MUMBAI	MAHARASHTRA	No Vulnerability Found in ECommerce Transaction	-779531e1f84f91f55c6245de36647f136361bcd4
140.205.32.3-10.42.0.211-443-55083-6	04-30-22	Shipped	Amazon.in	JNE3671	JNE3671-TU-XXXL	Top	3XL	1.0	574.0	Credit Card	CHENNAI	TAMIL NADU	No Vulnerability Found in ECommerce Transaction	-779531e1f84f91f55c6245de36647f136361bcd4
203.205.147.229-10.42.0.211-80-52309-6	04-30-22	Shipped	Amazon.in	SET264	SET264-KR-NP-XL	Set	XL	1.0	824.0	Debit card	GHAZIABAD	UTTAR PRADESH	No Vulnerability Found in ECommerce Transaction	-779531e1f84f91f55c6245de36647f136361bcd4
172.217.12.174-10.42.0.42-443-44753-6	04-30-22	Cancelled	Amazon.in	SET200	SET200-KR-NP-A-XXXL	Set	3XL	0.0	420.0	Credit Card	HYDERABAD	TELANGANA	No Vulnerability Found in ECommerce Transaction	-779531e1f84f91f55c6245de36647f136361bcd4
172.217.10.42-10.42.0.211-443-45534-6	04-30-22	Shipped	Amazon.in	JNE3160	JNE3160-KR-G-S	kurta	S	1.0	685.0	Credit Card	CHENNAI	TAMIL NADU	No Vulnerability Found in ECommerce Transaction	-779531e1f84f91f55c6245de36647f136361bcd4
10.42.0.211-64.71.142.124-	04-30-22	Shipped - Delivered	Amazon.in	JNE3405	JNE3405-KR-XS	kurta	XS	1.0	399.0	Credit Card	Amravati.	MAHARASHTRA	No Vulnerability Found in	-779531e1f84f91f55c6245de36647f136361bcd4

Fig : Vulnerability Not Found Transitions

Home User Server

Vulnerability Evaluation Method
for E-Commerce Transaction
Systems With Unobservable
Transitions



Find Datasets By Finite_State_Markov_Chain

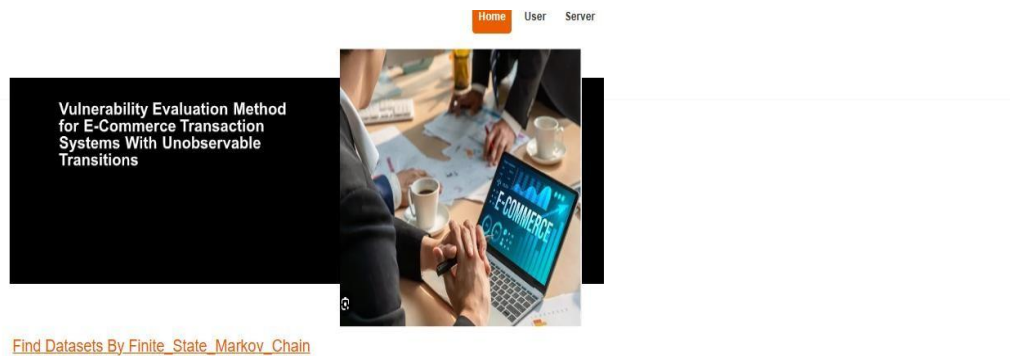
Select Evaluation Type

Vulnerability Found in ECommerce Transaction

Find Evaluation Type

[Back](#)

Fig : Finding Vulnerability Type



[Find Datasets By Finite State Markov Chain](#)

Order_ID	PDate	Status	Sales_Channel	Style	SKU	Category	PSize	Qty	Amount	payment_by	ship_city	ship_state	Evaluation	hashcode
182.22.31.252-10.42.0.211-443-48485-6	04-30-22	Cancelled	Amazon.in	SET389	SET389-KR-NP-S	Set	S	0.0	647.62	Credit Card	MUMBAI	MAHARASHTRA	Vulnerability Found in ECommerce Transaction	2a504c8ccea0b2a80ddf32d1742386b46e2c817
172.217.3.110-10.42.0.211-443-60421-6	04-30-22	Shipped - Delivered to Buyer	Amazon.in	JNE3781	JNE3781-KR-XXXL	kurta	3XL	1.0	406.0	Credit Card	BENGALURU	KARNATAKA	Vulnerability Found in ECommerce Transaction	2a504c8ccea0b2a80ddf32d1742386b46e2c817
172.217.12.206-10.42.0.42-443-35767-6	04-30-22	Cancelled	Amazon.in	J0341	J0341-DR-L	Western Dress	L	0.0	753.33	Credit Card	PUDUCHERRY	PUDUCHERRY	Vulnerability Found in ECommerce Transaction	2a504c8ccea0b2a80ddf32d1742386b46e2c817
10.42.0.151-91.190.216.54-54248-12350-6	04-30-22	Shipped	Amazon.in	J0095	J0095-SET-L	Set	L	1.0	653.0	Credit Card	CHANDIGARH	CHANDIGARH	Vulnerability Found in ECommerce Transaction	2a504c8ccea0b2a80ddf32d1742386b46e2c817
10.42.0.211-104.244.43.35-41821-443-6	04-30-22	Shipped - Delivered to Buyer	Amazon.in	JNE3405	JNE3405-KR-S	kurta	S	1.0	399.0	Credit Card	HYDERABAD	TELANGANA	Vulnerability Found in ECommerce Transaction	2a504c8ccea0b2a80ddf32d1742386b46e2c817
10.42.0.211-23.77.88.53-43881-443-6	04-30-22	Shipped	Amazon.in	JNE3461	JNE3461-KR-XXXL	kurta	XXXL	1.0	363.0	Credit Card	Chennai	TAMIL NADU	Vulnerability Found in ECommerce Transaction	2a504c8ccea0b2a80ddf32d1742386b46e2c817
10.42.0.1-...	04-30-				JNE3500-							UTTAR	Vulnerability Found in	

Fig : Vulnerability Not Found Transitions

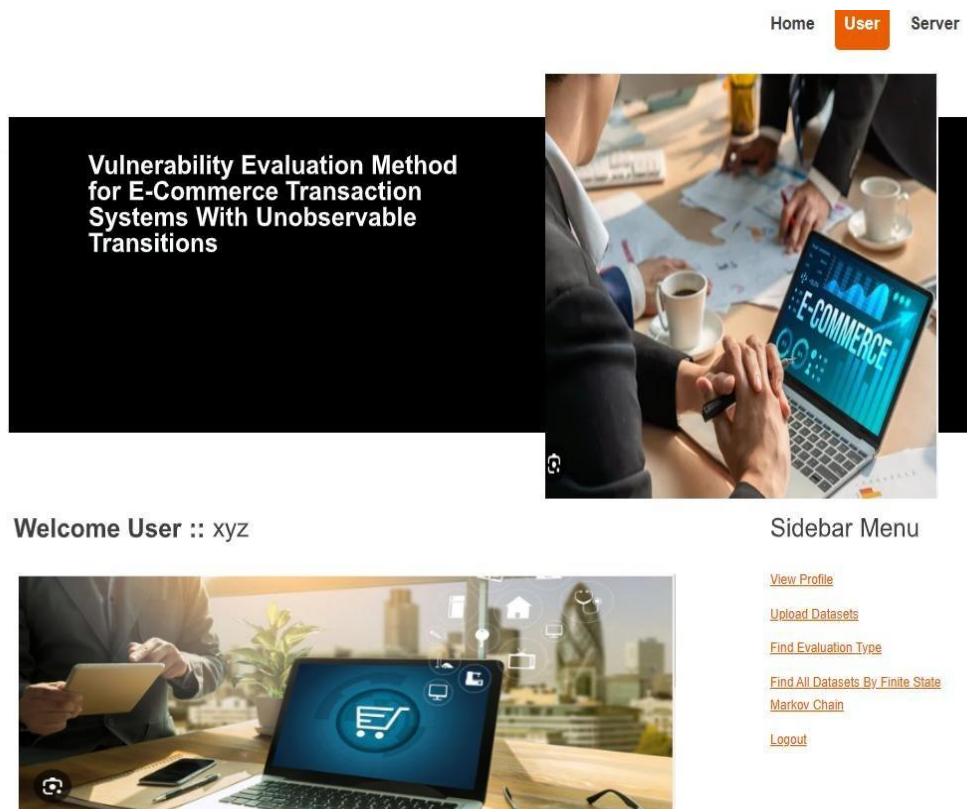


Fig : User Menu

CHAPTER 11

CONCLUSION

Due to the short times involved in online transactions and online payment platforms, online trading processes and trusted behavior issues are emerging along with the rapid development of online shopping and have gradually become a bottleneck in network trade development. Many e-commerce software systems are not mature and reliable, and they have flaws and mistakes that can be used by invaders. This leads to the emergence of security vulnerabilities and loss of user funds. This paper is motivated by the trusted behavior issues faced by these vulnerable network trade systems. Due to the uncomplicated graphical representation of a labeled Petri net, it can describe the overall structure of a local specification and unobservable actions well. However, the formal definition of its components can be used to provide precise abstraction. The VET-net is a subclass of labeled Petri nets. It can be used to model and simulate vulnerable e-commerce systems with unobservable actions. In this paper, on the basis of VET-nets, we describe the concept of vulnerable transitions, which include not only vulnerable actions but also unobservable transitions. Based on the concept of a slice, we then present a new method to locate the vulnerable transitions of a VET-net. We use hidden Markov models (HMMs) to approximate the state reach ability graph of a VET-net; this is called the HMM-RG method. Based on the HMM-RG, we describe the vulnerable state evaluation method of VET-nets. The vulnerable state evaluation method addresses the original problem. The proposed method is suitable for verifying and evaluating an online transaction system. It can also be used to verify simple e-commerce systems. The advantages of these methods are in dealing with unobservable actions. The proposed vulnerability assessment method can help designers analyze, diagnose and evaluate system vulnerabilities. Thus, the proposed method can be readily used in the system design and analysis of industrial online transaction business processes.

Due to neglecting data information, the methods are not suitable for some e-commerce systems, e.g., EBPN [4], [6], [13]. Regarding future work, there are other problems that require study related to the vulnerability evaluation of e-commerce systems, such as evaluation of data information and attack prevention.

CHAPTER 12

FUTURE SCOPE

We focus on improving the area could focus on enhancing the scalability and efficiency of the proposed vulnerability evaluation method. Additionally, exploring the integration of machine learning techniques to improve the detection accuracy of malicious states and transitions in e-commerce systems could be valuable. Moreover, investigating proactive measures for mitigating vulnerabilities identified through the evaluation process could further enhance the security of e-commerce transaction systems.

CHAPTER 13

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